A Nonparametric Eigenvalue-Regularized Integrated Covariance Matrix Estimator for Asset Return Data

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Abstract

The estimation of an integrated covariance matrix is important in finance for portfolio allocation or risk management. With the use of high frequency asset return data, such an estimator can be more adaptive to local volatility features, while sample size is significantly increased. To ameliorate the bias contributed from the extreme eigenvalues of the sample integrated covariance matrix when the dimension p of the matrix is large relative to the average daily sample size n, and the contamination by microstructure noise, various researchers attempted regularization with specific assumptions on the true matrix itself, like sparsity or factor structure, which can be restrictive at times. With non-synchronous trading and contamination of microstructure noise, we propose a nonparametrically eigenvalue-regularized integrated covariance matrix estimator (NERIVE) which does not assume specific structures for the underlying integrated covariance matrix. We show that NERIVE is almost surely positive definite, with extreme eigenvalues shrunk nonlinearly under the high dimensional framework $p/n \to c > 0$. We also prove that almost surely, the minimum variance optimal weight vector constructed using NERIVE has maximum exposure and actual risk upper bounds of order $p^{-1/2}$. Incidentally, the same maximum exposure bound is also satisfied by the theoretical minimum variance portfolio weights. All these results hold true also under a jump diffusion model for the log-price processes with jumps removed using the wavelet method proposed in Fan and Wang (2007). The practical performance of NERIVE is illustrated by comparing to the usual two-scale realized covariance matrix as well as some other nonparametric alternatives using different simulation settings and a real data set.

Key words and phrases. High frequency data; Microstructure noise; Non-synchronous trading; Integrated covariance matrix; Portfolio allocation; Nonlinear shrinkage.

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1 Introduction

In modern day finance, the so called tick-by-tick data on the prices of financial assets are readily available together with huge volume of other financial data. Advanced computational power and efficient data storage facilities mean that these data are analyzed on a daily basis by various market makers and academic researchers. While the Markowitz portfolio theory (Markowitz, 1952) is originally proposed for a finite number of assets using inter-day price data, the now easily accessible intra-day high frequency price data for a large number assets gives rise to new possibilities for efficient portfolio allocation, on top of the apparent increase in sample size for returns and volatility matrix estimation.

Certainly, the associated challenges for using high frequency data have to be overcome at the same time. One main challenge comes from the well documented market microstructure noise in the recorded tick-by-tick price data (Aït-Sahalia et al., 2005, Asparouhova et al., 2013). Another challenge comes from the non-synchronous trading times when more than one assets are considered. In terms of integrated covariance estimation, Xiu (2010) suggested a maximum likelihood approach for consistent estimation under market microstructure noise. Aït-Sahalia et al. (2010) proposed a quasi-maximum likelihood approach for estimating the covariance between two assets, while Zhang (2011) proposed a two or multi-scale covariance estimator to remove the bias accumulated due to the microstructure noise in the usual realized covariance formula, at the same time overcoming the non-synchronous trading times problem by using previous-tick times (see Section 2 also). Other attempts to overcome these two challenges together include Barndorff-Nielsen et al. (2011) and Griffin and Oomen (2011), to name but a few.

When there is more than one asset to manage, the integrated covariance matrix for the asset returns is an important input for risk management or portfolio allocation. A large number of assets requires an estimation of a large integrated covariance matrix. Even in the simplest case of independent and identically distributed random vectors, random matrix theory tells us that the sample covariance matrix will have severely biased extreme eigenvalues (see chapter 5.2 of Bai and Silverstein (2010) for instance). While high frequency data increases the sample size significantly, the problems of non-synchronized trading times across assets and contamination of market microstructure noise described in the previous paragraph make it even harder to obtain a satisfactory integrated covariance matrix when the number of assets has the same order as the intra-day sample size. To solve this, Wang and Zou (2010) assumes a sparsity condition and use thresholding to regularize a two or multi-scale covariance matrix estimator based on previous-tick times. Tao et al. (2011) uses this thresholded estimator to find a factor model structure for the daily dynamics of the integrated volatility matrix.

With respect to portfolio allocation, considering low frequency data, DeMiguel et al. (2009a) constraints the portfolio norm of a portfolio $\mathbf{w}$ using either the L_1 or squared L_2 norm, defined respectively by $\|\mathbf{w}\|_1 = \sum_i |w_i|$ and $\|\mathbf{w}\|_2^2 = \sum_i w_i^2$, and evaluate the performance of these portfolios empirically. With high frequency data, Fan et al. (2012) proposes to regularize the portfolio weights by constraining the L_1 norm of the portfolio, termed the gross exposure of a portfolio in the paper. They obtained remarkable theoretical and empirical results by using the pairwise refresh method to calculate individual elements in the two-scale covariance matrix. These two portfolio allocation methods do not require the regularization of the sample covariance matrix or the two (multi)-scale covariance matrix respectively (See Section 2.1 for the definition of a two-scale covariance matrix). The two-scale covariance matrix constructed in Fan et al. (2012) using the pairwise refresh method, however, may not be positive definite and adjustments are necessary to make it so again.

In this paper, we focus on the bias problem in the extreme eigenvalues of the two-scale covariance matrix, and propose to shrink them using a data splitting method for nonlinear shrinkage similar to Lam (2016). To give a simple demonstration of how serious the bias problem can be, suppose we have independent and identically distributed p-dimensional random vectors $\mathbf{X} = (x_1, \ldots, x_n)^{\mathrm{T}}$ with mean $\mathbf{0}$ and covariance matrix $\mathbf{\Sigma} = \sigma^2 \mathbf{I}_p$, where $\mathbf{I}_p$ is the $p \times p$ identity matrix. The Marčenko-Pastur Law (Marčenko and Pastur, 1967) states that the density function of the limiting spectrum of the sample covariance matrix $\mathbf{S} = n^{-1}\mathbf{X}\mathbf{X}^{\mathrm{T}}$ as $p, n \to$ with $p/n \to c > 0$, is

$$p_c(x) = \begin{cases} \frac{1}{2\pi x c \sigma^2}\sqrt{(b-x)(x-a)}, & a \leq x \leq b; \\ 0, & \text{otherwise}, \end{cases}$$

where $a = \sigma^2(1-\sqrt{c})^2$, $b = \sigma^2(1+\sqrt{c})^2$. See Bai and Silverstein (2009) Section 3.1 also. With this, say $p = 25$ and $n = 500$, i.e., p is just 5% of n, the largest and smallest eigenvalues are 50% larger and 40% smaller than the corresponding true ones respectively. It means that a seemingly small p is enough already for the sample covariance matrix to suffer from significant distortions for the extreme eigenvalues. When $\mathbf{\Sigma} \neq \sigma^2\mathbf{I}_p$ the distortion can potentially be more severe.

One of the main contribution from Lam (2016) is that it provides a possible way to perform nonlinear shrinkage on more than just independent and identically distributed random vectors, as in the case for the nonlinear shrinkage formula in Ledoit and Wolf (2012). Ultimately, we do not need to assume a particular structure of the underlying integrated covariance matrix, and can still show that regularization on the extreme eigenvalues is achieved in the two-scale covariance matrix, resulting in a positive definite integrated covariance matrix estimator. At the same time, using its inverse in the construction of the minimum variance portfolio induces a natural upper bound on the maximum exposure of the portfolio, which decays at a rate of $p^{-1/2}$, where p is the number of assets. The importance of this bound is that the theoretical minimum variance portfolio, which uses the true underlying integrated covariance matrix as an input, satisfies such a bound also. See Theorem 5 for more details. Our method also ensures the actual risk of the portfolio decays at a rate of $p^{-1/2}$ almost surely. These results highlight the usefulness of shrinkage of eigenvalues in the context of portfolio allocation, and the effectiveness of nonlinear shrinkage in portfolio allocation is fully demonstrated in Section 5.

The rest of the paper is organized as follows. Section 2 presents the notations and model for the high frequency data and introduce our way to perform nonlinear shrinkage on the two scale covariance matrix estimator. Asymptotic theories and detailed assumptions, including those involving jumps removed data in the case of jump-diffusion log-price processes, can be found in Section 3. Practical concerns and implementation can be found in Section 4, while all simulations and a thorough empirical study are presented in Section 5. We give the conclusion of the paper in Section 6, before all the proofs of the theorems in the paper in Section 7.

2 Framework and Methodology

Consider p assets with log price process $\mathbf{X}_t = (X_t^{(1)}, \ldots, X_t^{(p)})^{\mathrm{T}}$, which obeys the following diffusion process

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \boldsymbol{\sigma}_t d\mathbf{W}_t, \quad t \in [0,1], \tag{2.1}$$

so that the time period is normalized to have length 1. Denote L the integer representing the number of partitions of the data, so that

$$0 = \tau_0 < \tau_1 < \cdots < \tau_L = 1,$$

and $(\tau_{\ell-1}, \tau_\ell]$ represents the ℓth partition. The process $\mathbf{W}_t$ is a p-dimensional standard Brownian motion. The drift $\boldsymbol{\mu}_t \in \mathbb{R}^p$ and the volatility $\boldsymbol{\sigma}_t \in \mathbb{R}^{p\times p}$ are assumed to be continuous in t. For each time interval $[a,b] \subset [0,1]$, the corresponding integrated covariance matrix is defined as

$$\boldsymbol{\Sigma}(a,b) = \int_a^b \boldsymbol{\sigma}_u \boldsymbol{\sigma}_u^{\mathrm{T}} du.$$

This matrix is an important input in risk calculation of a portfolio and in Markowitz portfolio allocation. If we have a portfolio $\mathbf{w}$ which stays constant over a period of time $[a,b]$, then the risk of the portfolio over this period of time can be expressed as

$$R^{1/2}(\mathbf{w}) = (\mathbf{w}^{\mathrm{T}} \boldsymbol{\Sigma}(a,b) \mathbf{w})^{1/2} = \Big(\int_a^b \mathbf{w}^{\mathrm{T}} \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^{\mathrm{T}} \mathbf{w} \Big)^{1/2}.$$

The integrand $\mathbf{w}^{\mathrm{T}}\boldsymbol{\sigma}_t\boldsymbol{\sigma}_t^{\mathrm{T}}\mathbf{w}$ can be considered an instantaneous squared-risk at time t for $\mathbf{w}$, and hence $R(\mathbf{w})$ is a measure of the total risk accumulated over the period $[a, b]$. At the same time, in Markowitz portfolio allocation for instance, $\boldsymbol{\Sigma}(a, b)^{-1}$ is required for the construction of the minimum variance portfolio (see Section 3.2 for more details).

Let $\{v_s\}, 1 \leq s \leq nL$ be the set of all-refresh times for the log prices in $\mathbf{X}_t$, where $n(\ell)$ is the number of all-refresh times at partition ℓ, with $\ell = 1, \ldots, L$, and $n = L^{-1}\sum_{\ell=1}^{L} n(\ell)$ is the average number of all-refresh times in a partition. To recall, an all-refresh time v_s is the time when all assets have been traded at least once from the last all-refresh time v_{s-1}. Let $t_s^j \in (v_{s-1}, v_s]$ be the sth previous-tick time for the jth asset, which is the last trading time before or at v_s. For non-synchronous trading, $t_s^{j_1} \neq t_s^{j_2}$ for $j_1 \neq j_2$ in general. Also, high-frequency prices are typically contaminated by microstructure noise, so that at the all-refresh time v_s, we only observe

$$\mathbf{Y}(s) = \mathbf{X}(s) + \boldsymbol{\epsilon}(s), \; s = 1, \ldots, nL, \tag{2.2}$$

where $\mathbf{X}(s) = (X^{(1)}_{t_s^1}, \ldots, X^{(p)}_{t_s^p})^{\mathrm{T}}$, $\boldsymbol{\epsilon}(s) = (\epsilon^{(1)}_{t_s^1}, \ldots, \epsilon^{(p)}_{t_s^p})^{\mathrm{T}}$, with $\boldsymbol{\epsilon}(\cdot)$ independent of $\mathbf{X}(\cdot)$. Such independence is in line with Barndorff-Nielsen et al. (2011) and Fan et al. (2012), for instance. Each $\boldsymbol{\epsilon}(s)$ is independent of each other with mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_\epsilon$. While relaxation to serially dependent microstructure noise appeared in, e.g., Ait-Sahalia et al. (2011), such an assumption of independence in the microstructure noise is needed for our methodology to work, since in Lam (2016), we need partitions of data to be independent. Serial dependence in the microstructure noise can violate this assumption. This certainly leaves an important direction for future research work. Detailed assumptions of our paper can be found in Section 3.

2.1 Two-Scale Covariance Estimator

Contamination of microstructure noise in high-frequency data means that the usual realized covariance is heavily biased. Hence in Zhang (2011), a two-scale covariance estimator (TSCV) is introduced to remove this bias. In this paper, we use a slightly modified multivariate version of the two-scale covariance estimator, also by Zhang (2011). For $\ell = 1, \ldots, L$, define

$$\begin{aligned}
\langle \widehat{Y^{(i)}, Y^{(j)}} \rangle_\ell &= [Y^{(i)}, Y^{(j)}]_\ell^{(K)} - \frac{|S^\ell(K)|_K}{|S^\ell(1)|}[Y^{(i)}, Y^{(j)}]_\ell^{(1)}, \text{ with} \\
[Y^{(i)}, Y^{(j)}]_\ell^{(K)} &= \frac{1}{K}\sum_{r \in S^\ell(K)} (Y^{(i)}_{t_r^i} - Y^{(i)}_{t_{r-K}^i})(Y^{(j)}_{t_r^j} - Y^{(j)}_{t_{r-K}^j}), \quad \text{and} \\
S^\ell(K) &= \{r : t_r^i, t_{r-K}^i \in (\tau_{\ell-1}, \tau_\ell] \text{ for all } i\}, \; |S^\ell(K)|_K = \frac{|S^\ell(K)| - K + 1}{K}.
\end{aligned} \tag{2.3}$$

Note that $[Y^{(i)}, Y^{(j)}]_\ell^{(1)}$ is the usual realized covariance matrix when returns are calculated using adjacent previous-tick times, whereas $[Y^{(i)}, Y^{(j)}]_\ell^{(K)}$ can be seen as a realized covariance matrix when returns are calculated at time points which are K previous-tick times apart instead of 1 (so, another scale). Ultimately, while both are dominated by the market microstructure noise, the difference defined in $\widehat{\langle Y^{(i)}, Y^{(j)}\rangle}_\ell$ is proved in Zhang (2011) to be able to cancel out the dominating effect of the microstructure noise, and is defined as the two-scale covariance between the ith and the jth assets using all the all-refresh data points in the ℓth partition $(\tau_{\ell-1}, \tau_\ell]$. With this, and defining $\widehat{\langle X\rangle} = \widehat{\langle X, X\rangle}$, we define the two-scale covariance matrix (TSCV) for the partition $(\tau_{\ell-1}, \tau_\ell]$ to be

$$\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell) = \frac{1}{4}\left(\widehat{\langle Y^{(i)} + Y^{(j)}\rangle}_\ell^{(K)} - \widehat{\langle Y^{(i)} - Y^{(j)}\rangle}_\ell^{(K)}\right)_{1\le i,j\le p}. \tag{2.4}$$

We suppressed the dependence on K in the notation $\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell)$ and all related definitions in the next section. In Section 3, we show that K works well at the order $n^{2/3}$, which is indeed the order of magnitude suggested in Zhang (2011).

2.2 Our Proposed Integrated Covariance Matrix Estimator

Although the two-scale covariance estimator in (2.4) removes the bias contributed from the microstructure noise, it does not solve the bias issue for the extreme eigenvalues when p is large such that $p/n \to c > 0$. In order to regularize the integrated covariance matrix in the time period $(\tau_{j-1}, \tau_j]$, $j = 1, \ldots, L$, we follow Lam (2016) and consider a rotation-equivariant estimator $\mathbf{\Sigma}(\mathbf{D}) = \mathbf{P}_{-j}\mathbf{D}\mathbf{P}_{-j}$, where $\mathbf{D}$ is a diagonal matrix, and $\mathbf{P}_{-j}$ is orthogonal such that

$$\widetilde{\mathbf{\Sigma}}_{-j} = \mathbf{P}_{-j}\mathbf{D}_{-j}\mathbf{P}_{-j}^{\mathrm{T}}, \quad j = 1, \ldots, L, \quad \text{with } \widetilde{\mathbf{\Sigma}}_{-j} = \sum_{\ell\neq j} \widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell). \tag{2.5}$$

The use of $\mathbf{P}_{-j}$, instead of $\mathbf{P}$ which is the eigenmatrix of $\sum_\ell \widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell)$, is to allow for the independence between $\mathbf{P}_{-j}$ and $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$, which ultimately leads to regularization in the same spirit as Lam (2016). Independence comes from the diffusion model in (2.1), and the serial independence of microstructure noise assumed in (2.2).

With $\mathbf{P}_{-j}$, same as Lam (2016), we want to find a diagonal matrix $\mathbf{D}$ to solve

$$\min_{\mathbf{D}} \left\|\mathbf{P}_{-j}\mathbf{D}\mathbf{P}_{-j}^{\mathrm{T}} - \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\right\|_F, \tag{2.6}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. It is not difficult to show that $\mathbf{D} = \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})$ is the solution, where $\text{diag}(A)$ creates a diagonal matrix using the diagonal elements of A. Because of the independence between $\mathbf{P}_{-j}$ and $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$, ultimately, we can prove that (See Theorem 1) all the elements in $\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})$ are asymptotically almost surely close to those in $\mathbf{D} = \text{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})$. This allows us to define our integrated covariance matrix estimator for the partition $(\tau_{j-1}, \tau_j]$ to be

$$\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) = \mathbf{P}_{-j}\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})\mathbf{P}_{-j}^{\mathrm{T}}. \tag{2.7}$$

The overall integrated covariance matrix estimator for the period $[0, 1]$ is then defined to be

$$\widehat{\mathbf{\Sigma}}(0, 1) = \sum_{j=1}^{L} \widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) = \sum_{j=1}^{L} \mathbf{P}_{-j}\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})\mathbf{P}_{-j}^{\mathrm{T}}. \tag{2.8}$$

In using $\mathbf{P}_{-j}$, we assume that each interval $(\tau_{j-1}, \tau_j]$ is small, so that at the population level, the eigenvectors of each $\mathbf{\Sigma}_{-j} = \sum_{\ell \neq j} \mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)$ is very similar to those for $\mathbf{\Sigma}(0, 1)$. Then each $\mathbf{P}_{-j}$ should also be similar to the eigenmatrix $\mathbf{P}$ for $\widetilde{\mathbf{\Sigma}}(0, 1) = \sum_{\ell=1}^{L} \widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell)$. The estimator $\widehat{\mathbf{\Sigma}}(0, 1)$ in (2.8) is then

$$\begin{aligned}
\widehat{\mathbf{\Sigma}}(0, 1) &\approx \sum_{j=1}^{L} \mathbf{P}_{-j}\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})\mathbf{P}_{-j}^{\mathrm{T}} \\
&\approx \mathbf{P}\text{diag}\left(\mathbf{P}^{\mathrm{T}} \sum_{j=1}^{L} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}\right)\mathbf{P}^{\mathrm{T}} = \mathbf{P}\text{diag}(\mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}(0, 1)\mathbf{P})\mathbf{P}^{\mathrm{T}}.
\end{aligned}$$

The first approximation uses the result in Theorem 1 to be presented in Section 3 below. The estimator $\mathbf{P}\text{diag}(\mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}(0, 1)\mathbf{P})\mathbf{P}^{\mathrm{T}}$ can be considered an ideal rotation-equivariant estimator, where $\mathbf{P}$ is an eigenmatrix utilizing all of the all-refresh data points, and $\text{diag}(\mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}(0, 1)\mathbf{P})$ is the ideal diagonal matrix in terms of the Frobenius norm.

In practice, a small interval can be a trading day or a quarter of it, depending on the number of trading days of data we have and the number of all-refresh data points in them. In our simulations and empirical examples in Section 4, we use 5 or 1 day of training data with $(\tau_{\ell-1}, \tau_\ell]$ set at 1 day or a quarter of a day, with the number of all-refresh data points in the order of hundreds in each interval.

3 Asymptotic Theory

In this section, we show that our proposed estimator (2.7) in the jth partition of the data is asymptotically close to the corresponding ideal rotation-equivariant estimator

$$\mathbf{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j) = \mathbf{P}_{-j}\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})\mathbf{P}_{-j}^{\mathrm{T}}. \tag{3.1}$$

We compare to this estimator since it is also using $\mathbf{P}_{-j}$ as the eigenmatrix. However, $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ is replaced by the true integrated covariance matrix $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$. We first introduce some assumptions for our theorems to hold. In the following and hereafter, we denote $\lambda_{\min}(\cdot)$ the minimum eigenvalue of a square matrix, and $a \asymp b$ means that $a = O(b)$ and $b = O(a)$.

(A1) The drift process $\{\boldsymbol{\mu}_t\}_{t\in[0,1]}$ in model (2.1) is such that $\boldsymbol{\mu}_t = \mathbf{0}$ for all $t \in [0, 1]$.

(A2) The volatility process $\{\boldsymbol{\sigma}_u\}_{u\in[0,1]}$ satisfies

$$0 < C_1 \leq \min_{u\in[0,1]} \lambda_{\min}(\boldsymbol{\sigma}_u\boldsymbol{\sigma}_u^{\mathrm{T}}) \leq \max_{u\in[0,1]} \lambda_{\max}(\boldsymbol{\sigma}_u\boldsymbol{\sigma}_u^{\mathrm{T}}) \leq C_2 < \infty$$

uniformly as $p \to \infty$, where $C_1, C_2 > 0$ are generic constants.

(A3) The processes $\boldsymbol{\epsilon}(\cdot)$ and $\mathbf{X}(\cdot)$ defined in (2.2) are independent of each other. For fixed i, each $\epsilon^{(i)}_{t^i_s}$ is independent of each other for $s = 1, \ldots, nL$, and is normally distributed. The covariance matrix of $\boldsymbol{\epsilon}(s)$ is $\mathbf{\Sigma}_\epsilon$ for each s, which has uniformly bounded eigenvalues as $p \to \infty$.

(A4) The observation times are independent of $\mathbf{X}(\cdot)$ and $\boldsymbol{\epsilon}(\cdot)$, and the partition boundaries τ_ℓ, $\ell = 0, 1, \ldots, L$, satisfy $0 < C_3 \leq \min_{\ell=1,\ldots,L} L(\tau_\ell - \tau_{\ell-1}) \leq \max_{\ell=1,\ldots,L} L(\tau_\ell - \tau_{\ell-1}) \leq C_4 < \infty$, where C_3, C_4 are generic constant. Also, the all-refresh times v_s, $s = 1, \ldots, nL$ satisfy $\max_{s=1,\ldots,nL} nL(v_s - v_{s-1}) \leq C_5$ for a generic constant $C_5 > 0$. Moreover, $\max_{\ell=1,\ldots,L} L(\tau_\ell - v_{n(\ell)}) = o(1)$.

(A5) For the TSCV parameters, we set $J = 1$ and $K \asymp n^{2/3}$, with $L = O(n^{1/13})$.

Assumption (A1) is for simplicity of our proofs. It can be extended to $\{\boldsymbol{\mu}_t\}$ being locally bounded because of the way $[Y^{(i)}, Y^{(j)}]^{(K)}_\ell$ is defined. The independence parts in Assumption (A3) is seen in Fan et al. (2012) and also Barndorff-Nielsen et al. (2011), but we relax their assumptions to allow for correlated microstructure noise components. The normality assumption for $\boldsymbol{\epsilon}(s)$ in (A3) can be replaced by higher order moment assumptions.

The first part of Assumption (A4) is automatically satisfied if the boundary set $\{\tau_\ell\}_{0\leq\ell\leq L}$ is pre-set, for instance, to be the daily opening or closing time of the L days of data, or a quarter of it, just as described in Section 2.2. Assumptions (A2) and (A4) are there to control the size of the eigenvalues of $\mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)$ for each $\ell = 1, \ldots, L$.

In Assumption (A5), the parameter K in the TSCV is chosen to be exactly as what Zhang (2011) suggested. With this order of magnitude for K, we need L to be going to infinity with order as specified in (A5) in order to control the bias contributed from the microstructure noise. The order for K can actually be changed with an accompanying change in the order for L. We use $K \asymp n^{2/3}$ only just for the simplicity in the proof and ease of demonstration.

Theorem 1. *Let Assumptions (A1) to (A5) hold. For the all-refresh log-price data* $\mathbf{Y}(s)$, $s = 1, \ldots, nL$ *in (2.2), the integrated covariance matrix estimator constructed in (2.7) satisfies, as* $n, p \to \infty$ *such that* $p/n \to c > 0$,

$$\max_{j=1,\ldots,L} \left\| \widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) \mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p \right\| \xrightarrow{a.s.} 0,$$

where $\|\cdot\|$ *denotes the spectral norm of a matrix.*

The proof can be found in Section 7. Since $\mathbf{P}_{-j}$ is orthogonal, it is easy to see that $\mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_j)$ in (3.1) has

$$\mathrm{Cond}(\mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_{j-1})) \le \mathrm{Cond}(\mathbf{\Sigma}(\tau_{j-1}, \tau_j)),$$

where $\mathrm{Cond}(\cdot)$ is the condition number of a matrix, defined by dividing the maximum over the minimum magnitude of eigenvalue of the matrix. Theorem 1 then implies that almost surely, as $n, p \to \infty$ such that $p/n \to c > 0$,

$$\mathrm{Cond}(\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_{j-1})) \le \mathrm{Cond}(\mathbf{\Sigma}(\tau_{j-1}, \tau_j)).$$

This is the result of nonlinear shrinkage of eigenvalues of $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$. Incidentally, since all eigenvalues of $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$ are non-negative, the result of Theorem 1 also proves the following.

Corollary 2. *Let Assumptions (A1) to (A5) hold. Then as* $n, p \to \infty$ *such that* $p/n \to c > 0$, *the integrated covariance matrix estimator* $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ *in (2.7), and also* $\widehat{\mathbf{\Sigma}}(0, 1)$ *in (2.8), are almost surely positive definite as long as* $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$ *and* $\mathbf{\Sigma}(0, 1)$ *are.*

This corollary shows that the positive definiteness of an integrated covariance matrix is preserved in our proposed estimator almost surely as we have large enough sample size. In practice, like NERCOME introduced in Lam (2016) (see also Section 5 for simulation results in portfolio analysis), we always have positive definiteness of the estimator with a moderate sample size n and a similar dimension p.

3.1 Extension to Jump Diffusion Processes

Our method can be extended to include jumps in the underlying log-price process $\mathbf{X}_t$, as long as the jumps are removed first at a certain rate. We introduce the relevant model first. With jumps, the underlying log-price process is modeled as

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \boldsymbol{\sigma}_t d\mathbf{W}_t + d\mathbf{J}_t, \quad t \in [0, 1], \tag{3.2}$$

where $\boldsymbol{\mu}_t$ and $\boldsymbol{\sigma}_t$ are as in the pure diffusion model (2.1), and $\mathbf{J}_t = (J_t^{(1)}, \ldots, J_t^{(p)})^{\mathrm{T}}$ denotes a p dimensional right-continuous pure jump process. Each element in $\mathbf{J}_t$ is assumed to have finite activity in $[0, 1]$, so that there are only finite number of jumps in each log-price process $X_t^{(j)}$ in the time interval we consider. The $J_t^{(j)}$'s can be correlated with each other, and each is modeled by

$$J_t^{(j)} = \sum_{\ell=1}^{N_t^{(j)}} B_\ell^{(j)}, \quad t \in [0, 1],$$

where each count process $N_t^{(j)}$ can be correlated with each other. The same holds true for each jump size $B_\ell^{(j)}$. The quadratic covariation over $[0, 1]$ for the process $\mathbf{X}_t$ is then

$$QV = \int_0^1 \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^{\mathrm{T}} dt + \sum_{0 \le t \le 1} \Delta \mathbf{J}_t \Delta \mathbf{J}_t^{\mathrm{T}}, \tag{3.3}$$

where $\Delta \mathbf{J}_t = \mathbf{J}_t - \mathbf{J}_{t-}$. It is clear that an off-diagonal entry in $\Delta \mathbf{J}_t \Delta \mathbf{J}_t^{\mathrm{T}}$ will only be non-zero in general when both the corresponding log-price processes have jumps at the same time for at least once. It can correspond to, e.g., certain major market news reacted by a number of stocks at the same time. From (3.3), we see that the realized covariance matrix is contaminated by jumps and co-jumps among asset returns, on top of microstructure noise and bias of extreme eigenvalues when dimension is large relative to sample size.

We propose to use the wavelet method described in Section 3.2 of Fan and Wang (2007) to remove the jumps in the log-price processes and construct our nonlinear shrinkage estimator in (2.8) using the jumps-removed data. The wavelet approach is also considered in Xue et al. (2014) to test for the presence of jumps in high-frequency financial time series. We give more practical details on how we implement the wavelet method for each observed log-price process in Section 5.

Note that from Theorem 1 of Fan and Wang (2007), using our notations, we can deduce immediately that the finite number of jumps in each log-price process are removed at a rate at least $(nL)^{-1/4}$ using the wavelet method, with nL being the total number of all-refresh data points. Individual asset may do even better since we use all data points available in practice for each asset before evaluating the all-refresh time points. This jump removal rate is in fact the key to the successful adaptation of wavelet jumps removal to our proposed nonlinear shrinkage estimator. More detailed assumptions:

(W1) The wavelet used in jump estimation are differentiable.

(W2) For the jump part of $X_t^{(j)}$ in $[0, 1]$ for $j = 1, \ldots, p$, its jump locations $\eta_\ell^{(j)}$ and jump sizes $B_\ell^{(j)}$ satisfy

$$N_1^{(j)} < \infty, \; \eta_1^{(j)} < \ldots < \eta_\ell^{(j)} < \ldots, \; 0 < |B_\ell^{(j)}| < \infty \text{ almost surely.}$$

(W3) The processes $(\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t, \mathbf{W}_t)$, $(N_t^{(j)}, B_\ell^{(j)})$ and $\boldsymbol{\epsilon}_t$ are all independent of each other for all $j = 1, \ldots, p$.

These are technical assumptions adapted from Fan and Wang (2007). Assumption (W2) means that we are dealing with finite number of jumps for each log-price process, and the sizes of the jumps are bounded from 0 almost surely.

Theorem 3. *Let Assumptions (A1) to (A5) and (W1) to (W3) hold for the jump-diffusion model (3.2). Using the jumps-removed all-refresh log-price data $\mathbf{Y}^*(s)$, $s = 1, \ldots, nL$ in constructing the integrated covariance matrix estimator in (2.7), where $\mathbf{Y}^*(s)$ is similar to $\mathbf{Y}(s)$ defined in (2.2) for prices with no jumps, the same conclusions as in Theorem 1 and Corollary 2 hold.*

3.2 Application to Portfolio Allocation

Similar to Ledoit and Wolf (2012) and Lam (2016), instead of comparing the performance of $\widehat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ with $\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)$, we compare it with the ideal rotation-equivariant estimator $\boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)$ in (3.1), which replaces $\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ in (2.7) by the true integrated covariance matrix $\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)$. In terms of the optimal portfolio weights for the minimum-variance portfolio, defining $\mathbf{1}_p$ as a column vector of p ones, we define

$$\widehat{\mathbf{w}}_{\text{opt}} = \frac{\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}, \quad \mathbf{w}_{\text{Ideal}} = \frac{\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p},$$

where

$$\boldsymbol{\Sigma}_{\text{Ideal}}(0,1) = \sum_{j=1}^{L} \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j).$$

Since both are minimum variance portfolios over the period $[0,1]$, we naturally want to compare their associated (squared) risks, defined by

$$R(\mathbf{w}) = \mathbf{w}^{\mathrm{T}}\boldsymbol{\Sigma}(0,1)\mathbf{w}$$

for a particular portfolio weight vector $\mathbf{w}$. Define the efficiency of $\widehat{\mathbf{w}}_{\text{opt}}$ relative to $\mathbf{w}_{\text{Ideal}}$ by

$$\text{Eff}(\mathbf{w}_{\text{Ideal}}, \widehat{\mathbf{w}}_{\text{opt}}) = \frac{R(\mathbf{w}_{\text{Ideal}})}{R(\widehat{\mathbf{w}}_{\text{opt}})}. \tag{3.4}$$

With this definition, the efficiency is higher when $\widehat{\mathbf{w}}_{\text{opt}}$ has a smaller risk, and vice versa.

Theorem 4. *Let Assumptions (A1) to (A5) hold. Then as $n, p \to \infty$ such that $p/n \to c > 0$, we have $\text{Eff}(\mathbf{w}_{\text{Ideal}}, \widehat{\mathbf{w}}_{\text{opt}}) \xrightarrow{a.s.} 1$. The same conclusion holds under the jump-diffusion model (3.2) if Assumptions (W1) to (W3) hold also, and we are using the jumps-removed data as described in Section 3.1.*

The proof of this theorem is in Section 7. It says that the risk of the minimum variance portfolio constructed from using $\widehat{\mathbf{\Sigma}}(0,1)$ is asymptotically the same as that constructed using $\mathbf{\Sigma}_{\text{Ideal}}(0,1)$ almost surely. The practical performance of $\widehat{\mathbf{w}}_{\text{opt}}$ is illustrated in the empirical analysis in Section 5.

3.2.1 Maximum exposure and actual risk upper bounds

Unlike DeMiguel et al. (2009a) or Fan et al. (2012) which constraint the L_1 or L_2 norm of a portfolio vector $\mathbf{w}$ explicitly through a tuning parameter, our method enjoys a natural upper bound on the maximum exposure asymptotically almost surely. The maximum exposure of a portfolio vector $\mathbf{w}$ is defined as $\|\mathbf{w}\|_{\max} = \max_i |w_i|$. This bound is important since the theoretical minimum variance portfolio is also subjected to the same bound. At the same time, the actual risk $R^{1/2}(\widehat{\mathbf{w}}_{\text{opt}})$ also has a natural upper bound, as presented below.

Theorem 5. *Let Assumptions (A1) to (A5) hold. Define the theoretical minimum variance portfolio weight to be*

$$\mathbf{w}_{\text{theo}} = \frac{\mathbf{\Sigma}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\mathbf{\Sigma}(0,1)^{-1}\mathbf{1}_p}.$$

Then as $n, p \to \infty$ such that $p/n \to c > 0$, the maximum exposures of $\widehat{\mathbf{w}}_{\text{opt}}$ and $\mathbf{w}_{\text{theo}}$ satisfy almost surely

$$p^{1/2}\|\widehat{\mathbf{w}}_{\text{opt}}\|_{\max},\ p^{1/2}\|\mathbf{w}_{\text{theo}}\|_{\max} \leq \frac{\max_{1\leq j\leq L}\lambda_{\max}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}{\min_{1\leq j\leq L}\lambda_{\min}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}.$$

The actual risks of $\widehat{\mathbf{w}}_{\text{opt}}$ and $\mathbf{w}_{\text{theo}}$ satisfy almost surely

$$\begin{aligned}
p^{1/2}R^{1/2}(\widehat{\mathbf{w}}_{\text{opt}}) &\leq \frac{\max_{1\leq j\leq L}\lambda_{\max}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}{\min_{1\leq j\leq L}\lambda_{\min}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}\cdot\lambda_{\max}^{1/2}(\mathbf{\Sigma}(0,1)),\\
p^{1/2}R^{1/2}(\mathbf{w}_{\text{theo}}) &\leq \lambda_{\max}^{1/2}(\mathbf{\Sigma}(0,1)),
\end{aligned}$$

where $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the maximum and minimum eigenvalues of a matrix respectively. If Assumptions (W1) to (W3) hold also under the jump-diffusion model (3.2), then the same conclusions hold for the maximum exposure and actual risk bounds, as long as we are using the jumps-removed data as described in Section 3.1.

The proof of this theorem is in Section 7. The gross exposure constraint by Fan et al. (2012) or the L_2-norm constraint by DeMiguel et al. (2009a) are useful in constraining the total exposure of a portfolio and obtaining special ones like the no-short-sale portfolio (by setting $\|\mathbf{w}\|_1 \leq 1$). In practice, as illustrated by our simulation experiments and real data analysis in Section 5, the maximum exposure can still be large while these explicit constraints are in place. Certainly, there are a lot of examples where concentrated portfolios can be rewarding. However, with respect to the minimum variance portfolio, the theoretical one does satisfy an upper bound on the maximum exposure as presented in Theorem 5. Our method has the same almost sure upper bound which decays as p increases. As illustrated in Section 5, the maximum exposure in $\widehat{\mathbf{w}}_{\text{opt}}$ is on average smaller than other state-of-the-art methods in various settings, especially when using a quarter of a trading day as a partition. At the same time, in the real data analysis, the risk for our method, measured as the out-of-sample standard deviation of the return for a portfolio, is smaller than all other methods in the two portfolio studies. See Section 5 for more details.

4 Practical Implementation

There are two parameters that we can tune for potentially better performance, namely the partition $(\tau_{j-1}, \tau_j]$ of the period $[0, 1]$ (thus also determining L itself which represents the number of partitions), and the scale parameter K used in the TSCV in (2.4). For example, suppose we are given a period of 10 days of tick-by-tick data, if we set $(\tau_{j-1}, \tau_j]$ to be one day, then $L = 10$. Similar to the function $g(m)$ in equation (4.7) of Lam (2016), we propose to minimize the following criterion for a good choice of $\boldsymbol{\tau} = \{\tau_j\}_{0\leq j\leq L}$ and K:

$$g(\boldsymbol{\tau}, K) = \left\| \sum_{j=1}^{L} \left(\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) - \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) \right) \right\|_F^2, \tag{4.1}$$

where $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ and $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ are defined in (2.4) and (2.7) respectively. In practice, for a given L, we can divide $[0, 1]$ into equal partitions, so that $\boldsymbol{\tau}$ is then determined.

For the choice of K, since we are using $K \asymp n^{2/3}$ as in Zhang (2011), we can search $K = [bn^{2/3}]$ on a preset grid of constant b. In practice, we found from our simulation results and real data analysis that using $b = 1$ provide good results, and portfolio performance is not too different from using other values of b, hence in this paper we use $b = 1$.

5 Empirical Results

5.1 Simulation

In this section, we simulate high frequency trading transactions of 100 stocks for one year (250 trading days). Following Barndorff-Nielsen et al. (2011) and Fan et al. (2012), we simulate the price processes and the asynchronous transaction times independently. The observed log-price is defined as $X_t^{o(i)} = X_t^{(i)} + \varepsilon_t^{(i)}$, where $X_t^{(i)}$ represents the latent log-price, and the microstructure noise has $\varepsilon_t^{(i)} \overset{iid}{\sim} N(0, 0.0005^2)$. We generate $p = 100$ latent log-prices by the following multivariate factor model with stochastic volatilities:

$$dX_t^{(i)} = \mu^{(i)} dt + \sqrt{1 - (\rho^{(i)})^2} \sigma_t^{(i)} dB_t^{(i)} + \rho^{(i)} \sigma_t^{(i)} dW_t + C\nu^{(i)} dZ_t, \quad i = 1, ..., 100, \tag{5.1}$$

where $\{W_t\}$, $\{Z_t\}$ and the $\{B_t^{(i)}\}$'s are independent standard Brownian motions. The processes $\{W_t\}$ and $\{Z_t\}$ imitate factors in the market. The constant $C = \mathbf{1}_{\{\text{model 2}\}}$ is 0 for the first model we consider. We set $\rho^{(i)} = -0.7C$, so that it is 0 in the first model, and hence there are no factors. For the second model, $C = 1$, so that it contains two factors. The spot volatility $\sigma_t^{(i)} = exp(\varrho_t^{(i)})$ follows the independent Ornstein-Uhlenbeck process

$$d\varrho_t^{(i)} = \alpha^{(i)}(\beta_0^{(i)} - \varrho_t^{(i)})dt + \beta_1^{(i)} dU_t^{(i)},$$

where the $\{U_t^{(i)}\}$'s are independent standard Brownian motions. Other parameters of $X_t^{(i)}$ are set at $(\mu^{(i)}, \beta_0^{(i)}, \beta_1^{(i)}, \alpha^{(i)}) = (0.03x_1^{(i)}, -x_2^{(i)}, 0.75x_3^{(i)}, -1/40x_4^{(i)})$ and $\nu^{(i)} = exp(\beta_0^{(i)})$, where the $x_j^{(i)}$'s are independent uniform random variables on the interval $[0.7, 1.3]$. The initial value of each log-price is set at $X_0^{(i)} = 1$ and the starting spot volatility $\varrho_0^{(i)} = 0$.

For the transaction times, we generate 100 different Poisson processes with intensities $\lambda_1, ..., \lambda_{100}$ respectively. Since the normal trading time for one day is 23400 seconds, λ_i is set to be $0.01i \times 23400$, where $i = 1, ..., 100$.

5.2 Comparison of portfolio allocation performance

To compare the performance of different methods, we focus on the minimum variance portfolio

$$\mathbf{w}_{\text{opt}} = \frac{\mathbf{\Sigma}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\mathbf{\Sigma}(0,1)^{-1}\mathbf{1}_p}, \quad \text{which solves} \min_{\mathbf{w}:\mathbf{w}^{\mathrm{T}}\mathbf{1}_p=1} \mathbf{w}^{\mathrm{T}}\mathbf{\Sigma}(0,1)\mathbf{w}.$$

We first set the benchmark for comparisons. Following Fan et al. (2012), we create a theoretical portfolio $\mathbf{w}_{\text{theo}}$, which is a minimum variance portfolio with $\mathbf{\Sigma}(0,1)$ evaluated using the simulated latent log-prices at the finest grid (1 per second). For all other methods, we use the all-refresh time points evaluated from the data (we do not hold positions overnight for all methods to avoid overnight price jumps, since they are not what our study is about).

Other portfolios are constructed and compared to the theoretical minimum variance portfolio (THEO) above. The first one is the equal weight portfolio (EQUAL). The second one is the minimum variance portfolio with $\mathbf{\Sigma}(0,1)$ substituted by the two scale covariance matrix (TSCV). When $\mathbf{\Sigma}(0,1)$ is substituted with our estimator, we abbreviate it as NERIVE with one trading day as a partition length, and quarNERIVE when a partition length is a quarter of a trading day. We also compare with the gross exposure constraint

(GEC) method (Fan et al., 2012), and finally the L_2 norm constraint (NORM) (DeMiguel et al., 2009a) based on TSCV. We constructed 3 GEC portfolios with tuning parameters $c = 1, 2, 3$, as well as 3 NORM portfolios with tuning parameters $\delta = 0.1, 0.5, 1$ for comparisons. We do not use the pairwise refresh method for GEC to save significant computational time in both the simulations and the real data analysis, as well as that the features of our method can be compared more directly to those of GEC.

The portfolio exercise is carried out as follows for all methods. We invest 1 unit of capital to the different portfolios above at a certain start date (e.g., day 6 if we are using a 5-day training window), and rebalance the portfolio weights daily, moving the training window one day ahead. There are two investment strategies for comparisons under each model 1 or 2. The first one rebalances the portfolio daily with a 5-day training window. The second one rebalances the portfolio daily with a 1-day training window.

The quantities to be compared for different portfolios are as follows. For daily rebalancing with a k-day training window ($k = 1$ or 5), we calculate the annualized portfolio return and annualized out-of-sample standard deviation, given respectively by

$$\widehat{\mu} = 250 \times \frac{1}{250 - k} \sum_{i=k+1}^{250} \mathbf{w}^{\mathrm{T}} \mathbf{r}_i, \quad \widehat{\sigma} = \Big(250 \times \frac{1}{250 - k} \sum_{i=k+1}^{250} (\mathbf{w}^{\mathrm{T}} \mathbf{r}_i - \frac{\widehat{\mu}}{250})^2 \Big)^{1/2}.$$

The out-of-sample standard deviation is a good indicator of how much risk is associated with a portfolio (DeMiguel et al., 2009a). We also calculate the Sharpe ratio $\widehat{\mu}/\widehat{\sigma}$, and compare the average maximum exposure over the whole investment horizon, and the maximum of the maximum exposure over the same period. Since this is a simulation experiment, we can calculate the actual risk of a portfolio $\mathbf{w}$, $R^{1/2}(\mathbf{w}) = (\mathbf{w}^{\mathrm{T}} \boldsymbol{\Sigma} \mathbf{w})^{1/2}$, over a trading day. We compare the averaged actual risks of different methods over the whole investment horizon.

Table 1 shows the results for model (5.1) with no factors. The theoretical minimum variance portfolio is very similar to the equal weight portfolio, and hence both are having virtually the same actual risk and out-of-sample standard deviation, which are unsurprisingly the lowest among all methods. Apart from these two portfolios, NORM with $\delta = 0.1$ and GEC with $c = 1$ have the smallest of these two risk measures when we use a 5-day or 1-day training window respectively. Our proposed methods, especially quarNERIVE, have competitively small risks as well in both scenarios among all methods considered. Considering portfolio return as well, NERIVE and quarNERIVE both have decent returns and high Sharpe ratios, while NORM with $\delta = 1$ have among the largest Sharpe ratios in both scenarios. TSCV in fact has the highest return with a 5-day training window. However, it invests over 220% in a stock on average in each trading day, while selling hugely on others as well upon closer inspection. As shown in the table, it invests over 38 times of the capital in one stock in one of the trading day. It is then not surprising to see that the risks associated are huge, and hence it is not a very viable portfolio to follow. Portfolios using NERIVE or quarNERIVE usually have smaller maximum exposures on average compared with other methods, and have much less extreme buying or selling compared with TSCV, and even when compared with all GEC portfolios. NORM with $\delta = 0.1$ has among the smallest maximum exposures under various settings, which is not surprising since the restriction $\|\mathbf{w}\|_2^2 \leq 0.1$ essentially forces each portfolio weight to be small explicitly.

Table 2 shows the results for model (5.1) with factors. The out-of-sample standard deviations and actual risks of all methods have generally increased compared to the results in Table 1. While keeping decent out-of-sample standard deviations, our methods have among the highest Sharpe ratios no matter a 5-day or 1-day training window is used. TSCV has even wilder behaviour than under the first model with no factors, showing regularization is very important when the number of assets is comparable to the sample size, which is in the order of hundreds in our case.

Methods	Portfolio Return (%)	Out-of-Sample SD (%)	Sharpe Ratio	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	Actual Risk(%)
			daily rebalancing portfolio with 5-day training window			
THEO	-4.64	2.01	-2.31	1.20(0.03)	1.33	1.15
NERIVE	485.44	9.75	49.79	16.78(2.43)	25.00	6.71
quarNERIVE	413.89	9.18	45.09	14.20(2.03)	21.42	5.67
EQUAL	0.18	2.02	0.09	1.00(0.00)	1.00	1.15
TSCV	237.38	51.45	4.61	62.09(48.5)	457.79	31.64
GEC1	-5.97	11.28	-0.53	53.28(15.6)	97.23	7.14
GEC2	32.17	14.21	2.26	56.74(20.5)	119.84	8.61
GEC3	88.98	18.35	4.85	61.18(19.7)	134.30	10.35
NORM0.1	139.53	5.59	24.96	8.62(1.02)	11.62	3.52
NORM0.5	564.66	10.79	52.33	19.33(2.45)	26.99	7.83
NORM1	704.59	12.96	54.37	26.72(3.62)	39.93	10.61
			daily rebalancing portfolio with 1-day training window			
THEO	-4.87	2.01	-2.42	1.20(0.03)	1.33	1.15
quarNERIVE	544.35	4.97	109.53	8.59(1.22)	13.25	3.51
EQUAL	-0.01	2.02	0.00	1.00(0.00)	1.00	1.15
TSCV	874.37	218.58	4.00	220.57(426.0)	3824.23	191.63
GEC1	26.10	2.82	9.26	5.71(4.46)	46.35	1.63
GEC2	361.86	12.80	28.27	13.52(11.1)	94.01	3.44
GEC3	304.71	15.82	19.26	13.70(12.9)	129.28	3.81
NORM0.1	619.49	5.19	119.36	8.46(1.36)	12.47	3.48
NORM0.5	736.75	5.67	129.94	19.20(3.44)	29.10	4.35
NORM1	737.30	5.94	124.12	8.40(1.09)	12.06	4.60

Table 1: Simulation results for model 1 with no factors ($C = 0$ in (5.1)): Annualized portfolio return, annualized out-of-sample standard deviation, averaged maximum absolute weight (standard deviation in bracket), maximum of maximum absolute weight and actual risk for THEO, NERIVE, quarNERIVE, EQUAL, TSCV, GEC methods ($c = 1, 2, 3$), and NORM methods($\delta = 0.1, 0.5, 1$).

In addition to the measures in Table 1 and 2, we add some financial evaluations for comparisons in Table 3, including the Value at Risk (VaR), Return-Loss and the Certainty Equivalent Return (CEQ). The VaR we use here is the 95th percentile of the perceived risk in a trading day for a portfolio $\mathbf{w}$. That is, the VaR is defined as

$$\mathrm{VaR} = \Phi_{0.95}((\mathbf{w}^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}\mathbf{w})^{1/2}),$$

where $(\mathbf{w}^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}\mathbf{w})^{1/2}$ is the perceived risk of a portfolio $\mathbf{w}$, and $\Phi_{0.95}(X)$ denotes the 95th percentile of a

Methods	Portfolio Return (%)	Out-of-Sample SD (%)	Sharpe Ratio	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	Actual Risk(%)
			daily rebalancing portfolio with 5-day training window			
THEO	-8.06	10.10	-0.80	2.55(0.29)	3.56	7.45
NERIVE	490.45	18.85	26.02	38.60(6.27)	56.93	12.33
quarNERIVE	366.69	15.20	24.12	29.07(4.86)	42.83	10.42
EQUAL	-0.88	10.11	-0.09	1.00(0.00)	1.00	7.47
TSCV	4929.38	4353.60	1.13	475.32(5490.21)	86007.16	1684.21
GEC1	10.68	14.06	0.76	58.06(17.5)	96.84	9.34
GEC2	27.18	16.07	1.69	64.63(24.7)	123.39	9.96
GEC3	61.41	19.42	3.16	74.09(26.3)	155.72	11.23
NORM0.1	9.63	11.03	0.87	8.55(1.05)	12.27	7.87
NORM0.5	99.86	16.74	5.97	19.16(2.48)	28.27	9.44
NORM1	225.65	22.09	10.22	27.38(3.53)	38.71	11.07
			daily rebalancing portfolio with 1-day training window			
THEO	-7.32	10.14	-0.72	2.56(0.29)	3.56	7.45
quarNERIVE	734.32	12.73	57.68	21.41(3.27)	36.42	9.78
EQUAL	-0.20	10.14	-0.02	1.00(0.00)	1.00	7.47
TSCV	-1380.59	2509.95	-0.55	2068.07(23089.84)	364326.78	6394.60
GEC1	18.90	10.51	1.80	6.61(6.16)	43.29	7.53
GEC2	245.38	16.08	15.26	14.85(11.4)	102.67	7.87
GEC3	173.07	16.60	10.43	16.68(14.2)	159.69	8.02
NORM0.1	503.62	9.87	51.03	8.43(1.01)	11.30	7.86
NORM0.5	731.81	9.92	73.77	15.35(2.64)	24.75	8.79
NORM1	733.63	10.52	69.74	16.60(3.22)	27.60	9.00

Table 2: Simulation results for model 2 with factors ($C = 1$ in (5.1)): Annualized portfolio return, annualized out-of-sample standard deviation, averaged maximum absolute weight (standard deviation in bracket), maximum of maximum absolute weight and actual risk for THEO, NERIVE, quarNERIVE, EQUAL, TSCV, GEC methods ($c = 1, 2, 3$), and NORM methods($\delta = 0.1, 0.5, 1$).

random variable $\mathbf{X}$. The Return-Loss, which calculates the corresponding loss of return with respect to the equal weight portfolio EQUAL (DeMiguel et al., 2009b), is defined as

$$\text{return-loss} = \frac{\mu_{\text{ew}}}{\sigma_{\text{ew}}} \times \widehat{\sigma} - \widehat{\mu},$$

where μ_{ew} and σ_{ew} are respectively the return and out-of-sample standard deviation for the equal weight portfolio. Finally, as in DeMiguel et al. (2009b), the certainty equivalent return (CEQ), which calculates the return relative to the investors' risk aversion level, is defined as

$$\text{CEQ} = \widehat{\mu} - \frac{\nu}{2}\widehat{\sigma}^2,$$

where ν is the risk aversion. We choose $\nu = 5$ here, which is also investigated in DeMiguel et al. (2009b).

From Table 3, NERIVE and quarNERIVE have among the smallest VaR among all methods in all scenarios. The TSCV behaved badly as expected because of its wild behaviour observed from Table 1 and 2. Since EQUAL does not perform particularly well among other methods, the Return-Loss for most of the methods compared are negative, indicating better overall Sharpe ratio for other methods. The CEQ for NORM with $\delta = 1$ is the best in all scenarios except for using a 5-day training window when there are factors, where NERIVE performed the best.

	Without factors ($C = 0$ in (5.1))			With factors ($C = 1$ in (5.1))		
Methods	VaR (%)	Return-Loss (%)	CEQ (%)	VaR (%)	Return-Loss (%)	CEQ (%)
daily rebalancing portfolio with 5-day training window						
THEO	0.26	4.82	-4.74	1.23	7.18	-10.61
NERIVE	0.20	-484.57	483.07	0.91	-492.09	481.57
quarNERIVE	0.22	-413.07	411.78	0.76	-368.01	360.92
EQUAL	0.25	-	0.08	1.20	-	-3.43
TSCV	4.45	-232.80	171.20	64.15	-5308.33	-468915.71
GEC1	1.33	6.98	-9.15	1.64	-11.90	5.74
GEC2	1.47	-30.90	27.12	1.78	-28.58	20.73
GEC3	1.81	-87.34	80.56	2.03	-63.10	51.98
NORM0.1	0.25	-139.03	139.33	1.27	-10.59	6.59
NORM0.5	0.17	-563.70	564.19	1.60	-101.32	92.85
NORM1	0.19	-703.44	703.19	1.89	-227.57	213.45
daily rebalancing portfolio with 1-day training window						
THEO	0.27	4.86	-4.97	1.23	7.12	-9.89
quarNERIVE	0.01	-544.37	543.73	0.20	-734.57	730.27
EQUAL	0.25	-	-0.12	1.20	-	-2.77
TSCV	10.56	-875.45	-320.04	68.83	1331.08	-158876.44
GEC1	0.24	-26.11	25.90	1.17	-19.11	16.14
GEC2	0.55	-361.92	357.77	1.14	-245.70	238.92
GEC3	1.03	-304.79	298.46	1.40	-173.40	166.18
NORM0.1	0.00	-619.52	618.81	0.20	-503.81	501.18
NORM0.5	0.00	-736.78	735.95	0.08	-732.01	729.36
NORM1	0.01	-737.33	736.41	0.10	-733.84	730.86

Table 3: Simulation results: Financial evaluations. Value at Risk (VaR), Return-Loss and Certainty Equivalent Return (CEQ) for THEO, NERIVE, quarNERIVE, EQUAL, TSCV, GEC methods ($c = 1, 2, 3$), and NORM methods($\delta = 0.1, 0.5, 1$).

5.3 Portfolio allocation study

In this study, we choose the stocks based on two lists, the "Fifty Most Active Stocks on NYSE, Round Lots (mils. of shares), 2013" and "Fifty Most Active Stocks by Dollar Volume on NYSE ($ in mils.), 2013", from the New York Stock Exchange Data official website `http://www.nyxdata.com/`. There are 26 stocks appearing in both of the lists above, and 74 stocks in either of them. We downloaded all the trading transactions of these 74 stocks in Year 2013 from Wharton Research Data Services (WRDS, `https://wrds-web.wharton.upenn.edu/`). We omit the stock Sprint Corporation due to missing price data. We first clean all the data by the R-package "highfrequency", which follows the high frequency data cleaning steps presented in Barndorff-Nielsen et al. (2009). We conduct our portfolio allocation study on two portfolios, one with the $p = 26$ stocks appearing in both lists, and the other with $p = 73$ stocks appearing in either of the lists.

We carry out the same portfolio allocation exercises as in our simulations for both the 26-stock and 73-stock portfolios. First we do not remove jumps from the cleaned data. The results are displayed in Tables 4 and 5. Both NERIVE and quarNERIVE achieve the lowest out-of-sample standard deviations in the two scenarios presented for both portfolios, which are all under 10%. Because of this, the Sharpe ratio, return-loss ad CEQ for NERIVE and quarNERIVE are all consistently good relative to other methods, although not of the highest return among all methods. The VaR for our proposed methods are good as well

$\mathbf{p = 26}$ Method	Return (%)	Out-of-Sample SD(%)	Sharpe Ratio (%)	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	VaR (%)	Return Loss(%)	CEQ (%)
daily rebalancing portfolio with 5-day training window								
NERIVE	18.57	9.80	1.89	20.55(5.68)	41.18	1.03	2.11	16.16
quarNERIVE	20.99	9.69	2.17	19.67(4.95)	35.95	1.01	-0.54	18.64
EQUAL	24.25	11.49	2.11	3.85(0.00)	3.85	1.23	-	20.95
TSCV	16.91	13.52	1.25	42.50(13.0)	83.75	1.44	11.62	12.34
GEC1	28.01	10.99	2.55	30.48(11.2)	68.86	1.14	-4.82	24.99
GEC2	20.72	11.19	1.85	35.88(10.4)	81.24	1.20	2.90	17.59
GEC3	17.02	12.40	1.37	40.14(11.4)	82.57	1.36	9.15	13.17
NORM0.1	18.56	10.09	1.84	12.82(1.70)	18.77	1.09	2.74	16.02
NORM0.5	14.31	11.87	1.21	32.96(6.20)	51.88	1.32	10.74	10.79
NORM1	14.73	13.08	1.13	40.82(11.0)	74.26	1.42	12.88	10.45
daily rebalancing portfolio with 1-day training window								
quarNERIVE	17.81	9.97	1.79	18.64(6.74)	41.89	1.03	3.20	15.32
EQUAL	24.11	11.44	2.11	3.85(0.00)	3.85	1.22	-	20.84
TSCV	19.62	15.44	1.27	59.91(79.7)	972.52	1.59	12.92	13.65
GEC1	21.35	11.61	1.84	15.96(6.73)	55.98	1.23	3.12	17.98
GEC2	24.41	11.43	2.14	30.51(12.4)	72.09	1.17	-0.32	21.14
GEC3	27.56	12.22	2.26	36.95(14.2)	86.17	1.27	-1.81	23.83
NORM0.1	20.72	10.23	2.03	12.31(1.67)	19.12	1.08	0.84	18.10
NORM0.5	18.45	11.55	1.60	29.08(5.62)	47.11	1.24	5.89	15.12
NORM1	16.71	12.65	1.32	41.30(10.5)	70.33	1.39	9.95	12.71

Table 4: Empirical results for the 26 most actively traded stocks in NYSE: annualized portfolio return, annualized out-of-sample standard deviation, Sharpe ratio, averaged maximum absolute weight, maximum of maximum absolute weight, value at risk, return-loss and certainty equivalent return for NERIVE, quarNERIVE, EQUAL, TSCV, GEC with $c = 1, 2, 3$, and NORM with $\delta = 0.1, 0.5, \delta = 1$.

$\mathbf{p = 73}$ Method	Return (%)	Out-of-Sample SD(%)	Sharpe Ratio (%)	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	VaR (%)	Return Loss(%)	CEQ (%)
daily rebalancing portfolio with 5-day training window								
NERIVE	15.30	8.36	1.83	12.06(3.21)	21.93	0.89	0.40	13.55
quarNERIVE	16.12	8.66	1.86	12.08(3.09)	20.95	0.91	0.14	14.24
EQUAL	22.29	11.87	1.88	1.37(0.00)	1.37	1.28	-	18.77
TSCV	1367.06	1036.37	1.32	628.74(4042)	58950.10	25.02	579.08	-25484.57
GEC1	25.46	11.71	2.17	6.52(2.94)	22.33	1.26	-3.47	22.03
GEC2	22.57	11.22	2.01	11.45(7.45)	50.49	1.20	-1.50	19.43
GEC3	16.52	11.60	1.42	16.00(11.4)	104.48	1.29	5.26	13.16
NORM0.1	18.52	9.91	1.87	7.99(1.75)	13.27	1.09	0.09	16.06
NORM0.5	14.05	9.84	1.43	10.71(3.67)	23.48	1.11	4.43	11.63
NORM1	9.32	10.42	0.89	14.56(7.92)	48.36	1.20	10.25	6.60
daily rebalancing portfolio with 1-day training window								
quarNERIVE	18.76	9.26	2.03	8.49(2.73)	19.52	0.95	-1.23	16.62
EQUAL	22.42	11.84	1.89	1.37(0.00)	1.37	1.28	-	18.92
TSCV	-53.76	261.90	-0.21	373.58(670.7)	3896.86	15.20	549.69	-1768.51
GEC1	22.32	11.97	1.86	6.75(3.85)	31.67	1.30	0.35	18.74
GEC2	14.76	11.77	1.25	12.48(10.1)	88.36	1.31	7.53	11.30
GEC3	14.56	13.13	1.11	19.19(15.1)	106.75	1.39	10.30	10.25
NORM0.1	21.82	10.26	2.13	8.05(2.50)	18.52	1.07	-2.39	19.19
NORM0.5	31.63	17.60	1.80	18.21(10.4)	50.22	1.71	1.70	23.89
NORM1	33.67	24.83	1.36	27.27(16.7)	75.60	2.41	13.35	18.26

Table 5: Empirical results for the 73 most actively traded stocks in NYSE: annualized portfolio return, annualized out-of-sample standard deviation, Sharpe ratio, averaged maximum absolute weight, maximum of maximum absolute weight, value at risk, return-loss and certainty equivalent return for NERIVE, quarNERIVE, EQUAL, TSCV, GEC with $c = 1, 2, 3$, and NORM with $\delta = 0.1, 0.5, \delta = 1$.

compared to all other methods. The out-of-sample standard deviations are smaller for GEC and NORM with smaller values of constraints c and δ respectively in general.

Apart from the equal weight portfolio, the maximum absolute weight on average and the maximum of these are in general one of the smallest for NERIVE and quarNERIVE compared with other methods. Recall that the theoretical minimum variance portfolio has a natural upper bound the same as that for NERIVE and quarNERIVE almost surely from Theorem 5. The TSCV performs wildly again, and is more so when p is larger, showing that there is a more urgent need for regularization when p is larger.

We also considered jumps removed data. To follow the wavelet method in Section 3.1 to remove jumps as in (Fan and Wang, 2007), let $X_{j,k}$, $Y_{j,k}$ and $\epsilon_{j,k}$ be the wavelet coefficients of the original log-price series X_t, Y_t and ϵ_t, so that $Y_{j,k} = X_{j,k} + \epsilon_{j,k}, k = 1, ..., 2^j, j = 1, ..., \log_2(n)$. The resolution levels j_n are chosen to be $j_n = \log_2(2n/\log^2 n)$, and the threshold D_n is defined as $d\sqrt{2\log n}$, where d is the median of $\{|Y_{j_n,k}|, k = 1, ..., 2^{j_n}\}$ divided by 0.6745, a robust estimate of standard deviation. If the magnitude of a wavelet coefficient exceeds our defined threshold so that $|Y_{j_n,k} > D_n|$ for some k, we consider it as a jump and the corresponding jump location is estimated by $\widehat{\gamma} = k2^{-j_n}$. Denote $\widehat{q}$ as the estimated number of jumps in period $(\tau_{j-1}, \tau_j]$, and the estimated jump locations are denoted as $\widehat{\gamma}_1, ..., \widehat{\gamma}_{\widehat{q}}$. With each estimated jump location $\widehat{\gamma}_\ell$, a small window $[\widehat{\gamma}_\ell - \delta_n, \widehat{\gamma}_\ell + \delta_n]$ for some $\delta_n > 0$ is chosen. Following Fan and Wang (2007), we take δ_n as the square root of the total number of data points considered in a stock after data cleaning. We estimate the jump size as $\widehat{L}_\ell = \bar{Y}_{\widehat{\gamma}_\ell+} - \bar{Y}_{\widehat{\gamma}_\ell-}$, where $\bar{Y}_{\widehat{\gamma}_\ell+}$ and $\bar{Y}_{\widehat{\gamma}_\ell-}$ are the average values of Y_t over the period $[\widehat{\gamma}_\ell, \widehat{\gamma}_\ell + \delta_n]$ and $[\widehat{\gamma}_\ell - \delta_n, \widehat{\gamma}_\ell)$, respectively.

The results are presented in Table 6 and Table 7 for the jumps removed data. In general, the out-of-sample standard deviations of all methods improve slightly compared to the original data analysis, although not significantly different. This is due to the fact that the number of jumps estimated for each date is typically around 4 or 5, which is a very small number compared to the number of all-refresh data points. NERIVE and quarNERIVE still maintain the smallest out-of-sample standard deviations among all methods in all scenarios, with small accompanied VaR. GEC and NORM methods maintain higher returns in general for both portfolios, although the Sharpe ratios of NERIVE and quarNERIVE are still competitive because of the low out-of-sample standard deviations.

6 Conclusion

We generalize nonlinear shrinkage of eigenvalues in a large sample covariance matrix for independent and identically distributed random vectors (Lam, 2016) to that of a large two-scale covariance matrix estimator (TSCV) for high frequency returns, which are not identically distributed in general. To do this, we split

p = 26 Method	Return (%)	Out-of-Sample SD(%)	Sharpe Ratio	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	VaR (%)	Return Loss(%)	CEQ (%)
				daily rebalancing portfolio with 5-day training window				
NERIVE	17.06	9.81	1.74	20.42(6.10)	44.08	1.04	3.64	14.65
quarNERIVE	19.19	9.60	2.00	19.48(5.29)	33.59	1.01	1.07	16.89
EQUAL	24.25	11.49	2.11	3.85(0.00)	3.85	1.23	-	20.95
TSCV	16.54	12.92	1.28	41.45(12.9)	91.55	1.36	10.73	12.37
GEC1	29.86	10.99	2.72	30.22(10.9)	66.22	1.14	-6.67	26.84
GEC2	18.97	11.06	1.72	34.77(10.1)	80.41	1.18	4.37	15.91
GEC3	17.62	12.05	1.46	39.21(11.3)	87.91	1.28	7.81	14.00
NORM0.1	17.61	10.11	1.74	12.82(1.63)	18.14	1.09	3.73	15.06
NORM0.5	15.80	11.65	1.36	32.83(6.07)	52.08	1.26	8.79	12.41
NORM1	15.80	12.65	1.25	40.03(11.0)	75.68	1.34	10.90	11.80
				daily rebalancing portfolio with 1-day training window				
quarNERIVE	16.96	9.96	1.70	18.42(6.73)	41.20	1.02	4.03	14.48
EQUAL	24.11	11.44	2.11	3.85(0.00)	3.85	1.22	-	20.84
TSCV	23.42	14.40	1.63	55.35(48.9)	708.41	1.54	6.93	18.23
GEC1	29.22	10.76	2.72	28.39(12.4)	61.58	1.14	-6.54	26.33
GEC2	25.65	11.37	2.26	36.09(13.1)	83.54	1.16	-1.69	22.42
GEC3	21.19	12.40	1.71	42.40(15.1)	94.60	1.30	4.94	17.34
NORM0.1	20.90	12.21	1.71	12.30(1.68)	20.24	1.08	4.83	18.29
NORM0.5	16.58	11.54	1.44	29.54(6.02)	61.94	1.25	7.74	13.25
NORM1	17.94	12.59	1.42	41.51(10.3)	72.52	1.38	8.59	13.98

Table 6: Empirical results (jumps removed) for the 26 most actively traded stocks in NYSE: annualized portfolio return, annualized out-of-sample standard deviation, Sharpe ratio, averaged maximum absolute weight, maximum of maximum absolute weight, value at risk, return-loss and certainty equivalent return for NERIVE, quarNERIVE, EQUAL, TSCV, GEC with $c = 1, 2, 3$, and NORM with $\delta = 0.1, 0.5, \delta = 1$.

p = 73 Method	Return (%)	Out-of-Sample SD(%)	Sharpe Ratio	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	VaR (%)	Return Loss(%)	CEQ (%)
				daily rebalancing portfolio with 5-day training window				
NERIVE	14.94	8.53	1.75	11.72(3.33)	21.74	0.90	1.08	13.12
quarNERIVE	15.28	8.56	1.79	11.74(3.16)	21.14	0.89	0.79	13.45
EQUAL	22.29	11.87	1.88	1.37(0.00)	1.37	1.28	-	18.77
TSCV	-456.46	278.75	-1.64	235.69(502.9)	4693.48	14.48	979.91	-2398.99
GEC1	17.56	11.09	1.58	19.95(9.66)	53.16	1.21	3.27	14.49
GEC2	14.93	10.55	1.42	24.33(8.84)	64.03	1.13	4.88	12.15
GEC3	12.42	10.13	1.23	25.08(9.33)	60.80	1.15	6.60	9.85
NORM0.1	14.96	9.72	1.54	9.49(1.48)	15.99	1.07	3.29	12.60
NORM0.5	11.68	10.94	1.07	20.11(4.88)	41.24	1.22	8.86	8.69
NORM1	7.32	13.63	0.54	27.64(7.90)	61.49	1.51	18.28	2.68
				daily rebalancing portfolio with 1-day training window				
quarNERIVE	15.55	9.56	1.63	8.27(2.63)	17.62	1.01	2.55	13.26
EQUAL	22.42	11.84	1.89	1.37(0.00)	1.37	1.28	-	18.92
TSCV	271.32	189.92	1.43	365.55(838.9)	7964.54	12.18	88.31	-630.44
GEC1	20.36	11.50	1.77	7.75(12.4)	69.34	1.26	1.42	17.05
GEC2	22.59	13.26	1.70	20.68(13.1)	101.39	1.31	2.52	18.19
GEC3	20.10	21.90	0.92	32.68(27.6)	163.43	2.10	21.37	8.11
NORM0.1	21.57	10.31	2.09	8.22(2.47)	17.53	1.08	-2.05	18.92
NORM0.5	31.14	18.30	1.70	19.90(10.4)	50.98	1.85	3.51	22.77
NORM1	48.61	25.89	1.88	28.60(16.5)	74.19	2.50	0.41	31.85

Table 7: Empirical results (jumps removed) for the 73 most actively traded stocks in NYSE: annualized portfolio return, annualized out-of-sample standard deviation, Sharpe ratio, averaged maximum absolute weight, maximum of maximum absolute weight, value at risk, return-loss and certainty equivalent return for NERIVE, quarNERIVE, EQUAL, TSCV, GEC with $c = 1, 2, 3$, and NORM with $\delta = 0.1, 0.5, \delta = 1$.

the data into partitions and regularize the eigenvalues of the TSCV within a partition by the data from other partitions. Regularization is indeed achieved both theoretically and empirically, as demonstrated by the good performance in our simulations and portfolio allocation exercises.

While NERIVE works well in real data analysis, the assumption of independent market microstructure noise can still be stringent. Allowing for weak serial correlation is more realistic and can be a future research direction. Another very important generalization for future reseach is to allow for nonlinear shrinkage when data can be weakly dependent in general, which in itself can have a lot of usages outside of finance. Even within finance, allowing for conditional heteroscedasticity in the volatility can potentially allow for regularization of a conditionally forecasted covariance or integrated covariance matrix.

7 Proof of Theorems

We provide the proof of all the theorems of the paper in this section. Hereafter, we denote $\mathbf{A}\circ\mathbf{B} = (a_{ij}b_{ij})_{i,j}$, the Hadamard product between the matrices $\mathbf{A}$ and $\mathbf{B}$. Define

$$\mathbf{Z}^{\pm}(s) = \mathbf{Y}(s)\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\mathbf{Y}(s)^{\mathrm{T}}, \quad s = 1, \ldots, nL,$$

where $\mathbf{Y}(s)$ is defined in (2.2). We can then write

$$\mathbf{Z}^{\pm}(s) = \mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\mathbf{X}_{v_s}^{\mathrm{T}} + \boldsymbol{\epsilon}_s\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\boldsymbol{\epsilon}_s^{\mathrm{T}}, \tag{7.1}$$

where $\mathbf{X}_{v_s} = (X_{v_s}^{(1)}, \ldots, X_{v_s}^{(p)})^{\mathrm{T}}$ is the latent price of all assets at exactly time v_s, and

$$\boldsymbol{\epsilon}_s = \mathbf{Y}(s) - \mathbf{X}_{v_s} = \boldsymbol{\epsilon}(s) + \mathbf{X}(s) - \mathbf{X}_{v_s}.$$

We are now in position to define the multivariate versions of (2.3) and (2.4). For $\ell = 1, \ldots, L$, define

$$\begin{aligned} \widehat{\langle \mathbf{Z}^{\pm}, \mathbf{Z}^{\pm} \rangle}_\ell &= [\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(K)} - \frac{|S^\ell(K)|_K}{|S^\ell(1)|}[\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(1)}, \quad \text{with} \\ [\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(K)} &= \frac{1}{K}\sum_{s\in S^\ell(K)} (\mathbf{Z}^{\pm}(s) - \mathbf{Z}^{\pm}(s-K)) \circ (\mathbf{Z}^{\pm}(s) - \mathbf{Z}^{\pm}(s-K)), \end{aligned} \tag{7.2}$$

where $S^\ell(K)$ and $|S^\ell(K)|_K$ are as defined in (2.3). The TSCV $\widetilde{\boldsymbol{\Sigma}}(\tau_{\ell-1}, \tau_\ell)$ defined in (2.4) can then be written as

$$\widetilde{\boldsymbol{\Sigma}}(\tau_{\ell-1}, \tau_\ell) = \frac{1}{4}\left(\widehat{\langle \mathbf{Z}^{+}, \mathbf{Z}^{+} \rangle}_\ell - \widehat{\langle \mathbf{Z}^{-}, \mathbf{Z}^{-} \rangle}_\ell\right). \tag{7.3}$$

Define

$$\mathbf{X}_s^{\pm} = \mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\mathbf{X}_{v_s}^{\mathrm{T}}, \quad \mathbf{E}_s^{\pm} = \boldsymbol{\epsilon}_s\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\boldsymbol{\epsilon}_s^{\mathrm{T}},$$

so that $\mathbf{Z}^{\pm}(s)$ in (7.1) can be written as $\mathbf{Z}^{\pm}(s) = \mathbf{X}_s^{\pm} + \mathbf{E}_s^{\pm}$. Then for each positive integer K, we can write $[\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(K)}$ in (7.2) as

$$[\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(K)} = [\mathbf{X}^{\pm}, \mathbf{X}^{\pm}]_\ell^{(K)} + 2[\mathbf{X}^{\pm}, \mathbf{E}^{\pm}]_\ell^{(K)} + [\mathbf{E}^{\pm}, \mathbf{E}^{\pm}]_\ell^{(K)}, \tag{7.4}$$

where

$$[\mathbf{X}^{\pm}, \mathbf{E}^{\pm}]_\ell^{(K)} = \frac{1}{K}\sum_{s\in S^\ell(K)} (\mathbf{X}_s^{\pm} - \mathbf{X}_{s-K}^{\pm}) \circ (\mathbf{E}_s^{\pm} - \mathbf{E}_{s-K}^{\pm}),$$

and $[\mathbf{X}^{\pm}, \mathbf{X}^{\pm}]_\ell^{(K)}$ and $[\mathbf{E}^{\pm}, \mathbf{E}^{\pm}]_\ell^{(K)}$ are defined similarly. Hence we can decompose $\widetilde{\boldsymbol{\Sigma}}(\tau_{\ell-1}, \tau_\ell) = \sum_{i=1}^6 \mathbf{B}_i^\ell$, where

$$\begin{aligned}
\mathbf{B}_1^\ell &= \frac{1}{4}\left([\mathbf{X}^+, \mathbf{X}^+]_\ell^{(K)} - [\mathbf{X}^-, \mathbf{X}^-]_\ell^{(K)}\right), \\
\mathbf{B}_2^\ell &= -\frac{|S^\ell(K)|_K}{4|S^\ell(1)|}\left([\mathbf{X}^+, \mathbf{X}^+]_\ell^{(1)} - [\mathbf{X}^-, \mathbf{X}^-]_\ell^{(1)}\right), \\
\mathbf{B}_3^\ell &= \frac{1}{2}\left([\mathbf{X}^+, \mathbf{E}^+]_\ell^{(K)} - [\mathbf{X}^-, \mathbf{E}^-]_\ell^{(K)}\right), \\
\mathbf{B}_4^\ell &= -\frac{|S^\ell(K)|_K}{2|S^\ell(1)|}\left([\mathbf{X}^+, \mathbf{E}^+]_\ell^{(1)} - [\mathbf{X}^-, \mathbf{E}^-]_\ell^{(1)}\right), \\
\mathbf{B}_5^\ell &= \frac{1}{4}\left([\mathbf{E}^+, \mathbf{E}^+]_\ell^{(K)} - [\mathbf{E}^-, \mathbf{E}^-]_\ell^{(K)}\right), \\
\mathbf{B}_6^\ell &= -\frac{|S^\ell(K)|_K}{4|S^\ell(1)|}\left([\mathbf{E}^+, \mathbf{E}^+]_\ell^{(1)} - [\mathbf{E}^-, \mathbf{E}^-]_\ell^{(1)}\right).
\end{aligned} \tag{7.5}$$

We now present a series of Lemmas before proving Theorem 1.

Lemma 1. *Let Assumptions (A1) to (A6) hold. Then for unit vectors $\mathbf{x}_{ij}$ being independent of $\{\mathbf{X}_{v_s}\}_{s\in S^j(1)}$ for $i = 1, \ldots, p$ and $j = 1, \ldots, L$, we have for $\mathbf{B}_1^j$ defined in (7.5),*

$$\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}} - 1 \right| \xrightarrow{a.s.} 0.$$

At the same time, we have

$$\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_2^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}} \right| \xrightarrow{a.s.} 0.$$

Suppose in the definition of (7.4), $\mathbf{X}_{v_s}$ is replaced by

$$\mathbf{X}_{v_s} + \mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s},$$

where $\mathbf{J}_{v_s}$ *represents the accumulated jumps up to time* v_s *(see equation (3.2) and the details therein), and* $\widehat{\mathbf{J}}_{v_s}$ *represents the estimator of* $\mathbf{J}_{v_s}$ *using the wavelet method from Fan and Wang (2007) as described in Section 3.1. Then the same results in the lemma hold true if Assumptions (W1) to (W3) from Section 3.1 hold also.*

Proof of Lemma 1. We first prove the lemma without any jumps in the data. Define $\mathbf{D}_x$ as the diagonal matrix with the elements in a vector $\mathbf{x}$ as the diagonal, and that $\Delta_K \mathbf{X}_{v_s} = \mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}}$. Using the identity $\mathbf{x}^{\mathrm{T}}\mathbf{A} \circ \mathbf{B}\mathbf{y} = \mathrm{tr}(\mathbf{D}_x \mathbf{A} \mathbf{D}_y \mathbf{B}^{\mathrm{T}})$, with $\mathrm{tr}(\cdot)$ being the trace of a matrix, we have for any $\mathbf{x} \in \mathbb{R}^p$ and K a positive integer,

$$\begin{aligned}
\mathbf{x}^{\mathrm{T}}[\mathbf{X}^{\pm}, \mathbf{X}^{\pm}]_j^{(K)}\mathbf{x} &= \frac{1}{K}\sum_{s\in S^j(K)} \mathbf{x}^{\mathrm{T}}(\mathbf{X}_s^{\pm} - \mathbf{X}_{s-K}^{\pm}) \circ (\mathbf{X}_s^{\pm} - \mathbf{X}_{s-K}^{\pm})\mathbf{x} \\
&= \pm\frac{1}{K}\sum_{s\in S^j(K)} \mathrm{tr}\Big(\mathbf{D}_x(\Delta_K \mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p \Delta_K \mathbf{X}_{v_s}^{\mathrm{T}})\Big)^2 \\
&= \pm\frac{1}{K}\sum_{s\in S^j(K)} \mathrm{tr}\Big(\mathbf{D}_x \Delta_K \mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}}\mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}}\mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}} \\
&\qquad \pm \mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}}\mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} + \mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}}\mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}}\Big) \\
&= \pm\frac{2}{K}\sum_{s\in S^j(K)}\Big((\mathbf{x}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_s})^2 \pm \sum_i x_i \sum_i x_i(\Delta_K\mathbf{X}_{v_s})_i^2\Big). \qquad (7.6)
\end{aligned}$$

This implies that

$$\mathbf{x}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x} = \frac{1}{K}\sum_{s\in S^j(K)}(\mathbf{x}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_s})^2 = \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K}(\mathbf{x}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_{rK+\ell}})^2. \qquad (7.7)$$

In the above, $v_{rK+\ell} = v_{rK+\ell}^j$, where $\tau_{j-1} = v_0^j < v_1^j < \cdots < v_{n(j)}^j \le \tau_j$ are all the all-refresh time points in the partition $(\tau_{j-1}, \tau_j]$. Hereafter, the superscript j will be suppressed if there are no ambiguity arise for simpler notations. By the diffusion model (2.1), we can write

$$\Delta_K\mathbf{X}_{v_{rK+\ell}} = \mathbf{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})^{1/2}\mathbf{Z}_{r,\ell}^j, \qquad (7.8)$$

where the vectors $\mathbf{Z}_{r,\ell}^j$'s are independent of each other for different values of $r = 1, \ldots, |S^j(K)|_K$ and $j = 1, \ldots, L$. Each $\mathbf{Z}_{r,\ell}^j$ has independent entries each with mean 0 and variance 1. Hence, combining (7.7) and (7.8), we have

$$\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}} - 1\right| \le \max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} g_{r,\ell}^j\right| + R_1,$$

where $g^j_{r,\ell}$ and R_1 are defined as

$$g^j_{r,\ell} = \frac{(\mathbf{Z}^{j\mathrm{T}}_{r,\ell}\boldsymbol{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})^{1/2}\mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}, \tag{7.9}$$

$$R_1 = \max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\frac{1}{K}\sum_{\ell=0}^{K-1}\left(\sum_{r=1}^{|S^j(K)|_K}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})\mathbf{x}_{ij} - \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}\right)}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|. \tag{7.10}$$

To bound R_1, using Assumptions (A2) and (A4),

$$\begin{aligned}
R_1 &\le \frac{L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{\ell=0}^{K-1}\left(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_\ell, v_{|S^j(K)|-K+1+\ell})\mathbf{x}_{ij} - \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}\right)\right|\\
&= \frac{L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\sum_{\ell=0}^{K-1}\left(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1}, v_\ell)\mathbf{x}_{ij} + \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_{|S^j(K)|-K+1+\ell},\tau_j)\mathbf{x}_{ij}\right)\\
&\le \frac{L}{C_1C_3K}\sum_{\ell=0}^{K-1}\left(\frac{C_2C_5\ell}{nL} + \frac{C_2C_5(n(j)-|S^j(K)|+K-1-\ell)}{nL}\right) + \frac{C_2}{C_1C_3}o(1)\\
&= O(K/n) + o(1) = o(1),
\end{aligned}$$

since $|S^j(K)| = n(j) - K$. To bound the first term, we first note that $E(g^j_{r,\ell}|\mathbf{x}) = 0$. Using Lemma 2.7 of Bai and Silverstein (1998), for any integer $q \ge 2$,

$$\begin{aligned}
E(|g^j_{r,\ell}|^q|\mathbf{x}) &\le \frac{L^qK_q}{C_1^qC_3^q}\Big((E^{q/2}|z_1|^4 + E|z_1|^{2q})(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})\mathbf{x}_{ij})^q\Big)\\
&= O(K^qn^{-q}) = O(n^{-q/3}),
\end{aligned} \tag{7.11}$$

where we used the fact the all moments are finite for the components in $\mathbf{Z}^j_{r,\ell}$ since they are normally distributed by the assumption of the diffusion model (2.1), and we used Assumptions (A4) and (A6) in the last line. Then we have for $q \ge 1$,

$$\begin{aligned}
E\left(\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left(\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} g^j_{r,\ell}\right)^{2q}|\mathbf{x}\right) &\le E\left(\max_{\substack{i=1,\dots,p\\1\le j\le L}}\max_{1\le\ell\le K-1}\left(\sum_{r=1}^{|S^j(K)|_K} g^j_{r,\ell}\right)^{2q}|\mathbf{x}\right)\\
&= O\Big(pLK\sum_{r_1\ne\cdots\ne r_q}\prod_{k=1}^{q}E((g^j_{r_k,\ell})^2|\mathbf{x})\Big)\\
&= O(pLK\cdot|S^j(K)|_K^q\cdot n^{-2q/3}) = O(n^{-q/3+68/39}),
\end{aligned}$$

where we used the fact that the largest order from the expansion in the first line must comes from the product of all square terms. Hence we can set q with $q \ge 9$, so that $\sum_{n\ge1} n^{-q/3+68/39} < \infty$, proving that $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} g^j_{r,\ell}\right| \xrightarrow{a.s.} 0$ through the Borel-Cantelli lemma. This proves the first part of

Lemma 1.

For the second part, applying (7.6) and (7.7) with $K = 1$, we have

$$\max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}} \mathbf{B}_2^j \mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{x}_{ij}} \right| \leq \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \frac{|S^j(K)|_K}{|S^j(1)|} \sum_{r=1}^{S^j(1)} g_{r,0}^j \right| + R_2,$$

where $g_{r,0}^j$ is defined as in (7.9) with $K = 1$, and R_2 is defined as and bounded by

$$R_2 = \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \frac{L|S^j(K)|_K}{C_1C_3|S^j(1)|} \left| \sum_{r=1}^{|S^j(1)|} \mathbf{x}_{ij}^{\mathrm{T}} \mathbf{\Sigma}(v_{r-1}, v_r) \mathbf{x}_{ij} \right| \leq \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \frac{L|S^j(K)|_K}{C_1C_3|S^j(1)|} \frac{C_2C_4}{L} = O(K^{-1}) = o(1).$$

Finally, $E(|g_{r,0}^j|^q|\mathbf{x}) = O(n^{-q})$ for $q \geq 2$ by (7.11), and

$$E\Big(\max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big(\frac{|S^j(K)|_K}{|S^j(1)|} \sum_{r=1}^{|S^j(1)|} g_{r,0}^j \Big)^{2q} \Big) = O(pLK^{-2q}|S^j(1)|^q n^{-2q}) = O(n^{-7q/3+14/13}),$$

so that $q = 1$ here is sufficient for showing $\max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \frac{|S^j(K)|_K}{|S^j(1)|} \sum_{r=1}^{S^j(1)} g_{r,0}^j \right| \overset{a.s.}{\to} 0$.

We now prove the results when $\mathbf{X}_{v_s}$ is replaced by $\mathbf{X}_{v_s} + \mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s}$. To this end, by the arguments presented in Section 3.1, with Assumptions (W1) to (W3) in place, we have the elementwise error rate

$$\Delta_K \mathbf{J}_{v_s} - \Delta_K \widehat{\mathbf{J}}_{v_s} = (\mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s}) - (\mathbf{J}_{v_{s-K}} - \widehat{\mathbf{J}}_{v_{s-K}}) = O_P((nL)^{-1/4}),$$

so that for an independent unit vector $\mathbf{x}$, we still have $\mathbf{x}^{\mathrm{T}}(\Delta_K \mathbf{J}_{v_s} - \Delta_K \widehat{\mathbf{J}}_{v_s}) = O_P((nL)^{-1/4})$. Then by (7.7), we have by defining $\mathbf{x}^{\mathrm{T}}\mathbf{B}_1^{*j}\mathbf{x}$ to be the original $\mathbf{x}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x}$ with $\mathbf{X}_{v_s}$ replaced by $\mathbf{X}_{v_s} + \mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s}$,

$$\begin{aligned}
\mathbf{x}^{\mathrm{T}}\mathbf{B}_1^{*j}\mathbf{x} &= \frac{1}{K} \sum_{s\in S^j(K)} (\mathbf{x}^{\mathrm{T}}\Delta_K \mathbf{X}_{v_s} + \mathbf{x}^{\mathrm{T}}(\Delta_K \mathbf{J}_{v_s} - \Delta_K \widehat{\mathbf{J}}_{v_s}))^2 = \mathbf{x}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x} + A_1 + A_2, \quad \text{where} \\
A_1 &= \frac{2}{K} \sum_{\ell=0}^{K-1} \sum_{r=1}^{|S^j(K)|_K} \mathbf{x}^{\mathrm{T}}\Delta_K \mathbf{X}_{v_{rK+\ell}} \cdot \mathbf{x}^{\mathrm{T}}(\Delta_K \mathbf{J}_{v_{rK+\ell}} - \Delta_K \widehat{\mathbf{J}}_{v_{rK+\ell}}), \\
A_2 &= \frac{1}{K} \sum_{\ell=0}^{K-1} \sum_{r=1}^{|S^j(K)|_K} (\mathbf{x}^{\mathrm{T}}(\Delta_K \mathbf{J}_{v_{rK+\ell}} - \Delta_K \widehat{\mathbf{J}}_{v_{rK+\ell}}))^2.
\end{aligned}$$

Since by Assumption (W2) we have only finite number of jumps, the inner summation for A_1 and A_2 contain only constant number of non-zero terms, say $r = r_1, r_2, \ldots, r_m$ with m being a constant. Then defining $A_{1,ij}$ and $A_{2,ij}$ to be A_1 and A_2 with $\mathbf{x}$ replaced by independent unit vectors $\mathbf{x}_{ij}$ respectively, with

C a generic constant,

$$
\begin{aligned}
E\left(\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{A_{1,ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\right)^8 &\le pL\cdot CL^8E|A_{1,ij}|^8\\
&\le CpL^9E^{1/2}\left(\max_{0\le\ell\le K-1}\max_{r=r_1,\ldots,r_m}|\mathbf{x}_{ij}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_{rK+\ell}}|^{16}\right)\\
&\cdot E^{1/2}\left(\max_{0\le\ell\le K-1}\max_{r=r_1,\ldots,r_m}|\mathbf{x}_{ij}^{\mathrm{T}}(\Delta_K\mathbf{J}_{v_{rK+\ell}}-\Delta_K\widehat{\mathbf{J}}_{v_{rK+\ell}})|^{16}\right)\\
&\le mpKL^9O(K^4(nL)^{-4})O(n^{-2}L^{-2})=O(n^{-56/39}),
\end{aligned}
$$

so that $\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{A_{1,ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0$ because $\sum_{n\ge1}n^{-56/39}<\infty$. In the above arguments, we used two results. One is that by (7.8) and Lemma 2.7 of Bai and Silverstein (1998),

$$
E|\mathbf{x}_{ij}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_{rK+\ell}}|^{16}=O(K^8(nL)^{-8}).
$$

Another one is the following. Since $\widehat{\mathbf{J}}_{v_s}$ is constructed from the a linear combination of the return data and the normally distributed microstructure noises, and since for each $j=1,\ldots,p$ there are finite number of jumps $B_\ell^{(j)}$'s which are finite themselves, we have that $\mathbf{x}_{ij}^{\mathrm{T}}(\Delta_K\mathbf{J}_{v_{rK+\ell}}-\Delta_K\widehat{\mathbf{J}}_{v_{rk+\ell}})$ has moments of all order, and so

$$
E|\mathbf{x}_{ij}^{\mathrm{T}}(\Delta_K\mathbf{J}_{v_{rK+\ell}}-\Delta_K\widehat{\mathbf{J}}_{v_{rk+\ell}})|^{16}=O((nL)^{-1/4\cdot16})=O(n^{-4}L^{-4}).
$$

Similar to the arguments above, we can prove that

$$
\begin{aligned}
E\left(\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{A_{2,ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\right)^8 &\le CpL^9E\left(\max_{0\le\ell\le K-1}\max_{r=r_1,\ldots,r_m}|\mathbf{x}_{ij}^{\mathrm{T}}(\Delta_K\mathbf{J}_{v_{rK+\ell}}-\Delta_K\widehat{\mathbf{J}}_{v_{rK+\ell}})|^{16}\right)\\
&=O(pKL^9\cdot n^{-4}L^{-4})=O(n^{-76/39}),
\end{aligned}
$$

so that $\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{A_{2,ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0$ since $\sum_{n\ge1}n^{-76/39}<\infty$.

Together with the above results for $A_{1,ij}$ and $A_{2,ij}$, we have

$$
\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_1^{*j}\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}-1\right|\xrightarrow{a.s.}0.
$$

Using very similar arguments, we can show the respective result for $\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_2^{*j}\mathbf{x}_{ij}$. This completes the proof of the lemma. $\square$

Lemma 2. *Let Assumptions (A1) to (A6) hold. Then for unit vectors* $\mathbf{x}_{ij}$ *being independent of* $\{\mathbf{X}_{v_s}\}_{s\in S^j(1)}$ *and* $\{\boldsymbol{\epsilon}_s\}_{s\in S^j(1)}$ *for* $i = 1, \ldots, p$ *and* $j = 1, \ldots, L$, *we have for* $\mathbf{B}_3^j, \mathbf{B}_4^j$ *defined in (7.5),*

$$\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{B}_3^j + \mathbf{B}_4^j)\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}} \right| \xrightarrow{a.s.} 0.$$

Suppose in the definition of (7.4), $\mathbf{X}_{v_s}$ *is replaced by*

$$\mathbf{X}_{v_s} + \mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s},$$

where $\mathbf{J}_{v_s}$ *represents the accumulated jumps up to time* v_s *(see equation (3.2) and the details therein), and* $\widehat{\mathbf{J}}_{v_s}$ *represents the estimator of* $\mathbf{J}_{v_s}$ *using the wavelet method from Fan and Wang (2007) as described in Section 3.1. Then the same results in the lemma hold true if Assumptions (W1) to (W3) from Section 3.1 hold also.*

Proof of Lemma 2. Similar to the arguments in (7.6), we can show that

$$\begin{aligned}
&\frac{1}{2}(\mathbf{x}_{ij}^{\mathrm{T}}[\mathbf{Z}^+,\mathbf{E}^+]_j^{(K)}\mathbf{x}_{ij} - \mathbf{x}_{ij}^{\mathrm{T}}[\mathbf{Z}^-,\mathbf{E}^-]_j^{(K)}\mathbf{x}_{ij}) = \frac{2}{K}\sum_{s\in S^j(K)} (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\boldsymbol{\epsilon}_s - \boldsymbol{\epsilon}_{s-K})\\
&= \frac{2}{K}\sum_{s\in S^j(K)} (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s-K) - (\mathbf{X}_{v_s} - \mathbf{X}(s)) + (\mathbf{X}_{v_{s-K}} - \mathbf{X}(s-K))).
\end{aligned}$$

Hence we can decompose $\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_3^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}} \right| \le R_\epsilon - R_\epsilon^{(K)} - R_x + R_x^{(K)}$, where

$$\begin{aligned}
R_\epsilon &= \frac{2L}{C_1C_3K}\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \sum_{s\in S^j(K)} (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s) \right|,\\
R_\epsilon^{(K)} &= \frac{2L}{C_1C_3K}\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \sum_{s\in S^j(K)} (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s-K) \right|,\\
R_x &= \frac{2L}{C_1C_3K}\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \sum_{s\in S^j(K)} (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v_s} - \mathbf{X}(s)) \right|,\\
R_x^{(K)} &= \frac{2L}{C_1C_3K}\max_{\substack{i=1,\ldots,p\\1\le j\le L}} \left| \sum_{s\in S^j(K)} (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v_{s-K}} - \mathbf{X}(s-K)) \right|.
\end{aligned}$$

Define, for $s \in S^j(K)$, $g_{j,s} = (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s)$. Then since $\boldsymbol{\epsilon}(\cdot)$ is independent of $\mathbf{X}(\cdot)$, we have

$E(g_{j,s}|\mathbf{x}) = 0$ and the $g_{j,s}|\mathbf{x}$'s are uncorrelated with each other for fixed j. We also have for $q \geq 2$,

$$\begin{aligned}
E(|g_{j,s}|^q|\mathbf{x}) &\leq E^{1/2}\big((\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}|\mathbf{x}\big)^{2q} E^{1/2}\big(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s)\big|\mathbf{x}\big)^{2q} \\
&= E^{1/2}\Big(\mathbf{Z}_s^{j\mathrm{T}}\boldsymbol{\Sigma}(v_{s-K}, v_s)^{1/2}\mathbf{x}_{ij}|\mathbf{x}\Big)^{2q} E^{1/2}\big(\mathbf{Z}_{\epsilon,s}^{j\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij}|\mathbf{x}\big)^{2q} \\
&= O\big((\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_{s-K}, v_s)\mathbf{x}_{ij})^{q/2}(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon\mathbf{x}_{ij})^{q/2}\big) = O(K^{q/2}n^{-q/2}L^{-q/2}),
\end{aligned}$$

where the $\mathbf{Z}_s^j$ and $\mathbf{Z}_{\epsilon,s}^j$ are independent of each other for each s, with independent normal entries by the diffusion model (2.1) and Assumption (A3). With this, for integer $q \geq 1$,

$$\begin{aligned}
E|R_\epsilon|^{2q} &= O\bigg(L^{2q}K^{-2q} \cdot pL \cdot \Big(\sum_{s \in S^j(K)} g_{j,s}\Big)^{2q}\bigg) = O(pL^{2q+1}K^{-2q}|S^j(K)|^q \cdot K^q n^{-q}L^{-q}) \\
&= O(pL^{q+1}K^{-q}) = O(n^{-23q/39+14/13}),
\end{aligned}$$

and we can set $q = 3$ so that $\sum_{n \geq 1} n^{-23q/39+14/13} < \infty$, proving via the Borel-Cantelli lemma that $R_\epsilon \xrightarrow{a.s.} 0$. Similar arguments show that $R_\epsilon^{(K)} \xrightarrow{a.s.} 0$ also.

Consider $R_x \leq R_{x,1} + R_{x,2}$, where

$$\begin{aligned}
R_{x,1} &= \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1 \leq j \leq L}} \Big|\sum_{s \in S^j(K)} ((\mathbf{X}_{v_s} - \mathbf{X}(s))^{\mathrm{T}}\mathbf{x}_{ij})^2\Big|, \\
R_{x,2} &= \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1 \leq j \leq L}} \Big|\sum_{s \in S^j(K)} (\mathbf{X}(s) - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v_s} - \mathbf{X}(s))\Big|.
\end{aligned}$$

To analyze both terms, consider the previous-tick time $t_s^i \in (v_{s-1}, v_s]$ for the ith asset, which should satisfies

$$v_{s-1} < t_s^{(i_1)} \leq t_s^{(i_2)} \leq \cdots \leq t_s^{(i_p)} = v_s,$$

where $i_1, \ldots, i_p$ is some permutation of $1, \ldots, p$. Letting b_s denotes the number of tides, we can write the above as

$$v_{s-1} < t_s^{j_1} < t_s^{j_2} < \cdots < t_s^{j_{p-b_s}} = v_s,$$

where $j_1, \ldots, j_{p-b_s} \in \{1, \ldots, p\}$.

Then we can write, for $s = 1, \ldots, nL$,

$$\mathbf{X}_{v_s} - \mathbf{X}(s) = \sum_{k=1}^{p-b_s-1} \mathbf{D}_k^s \boldsymbol{\Sigma}(t_s^{j_k}, t_s^{j_{k+1}})^{1/2}\mathbf{Z}_k^s, \tag{7.12}$$

where $\mathbf{D}_k^s$ is a diagonal matrix with either 0 or 1 as elements. The jth diagonal element is 1 if the jth asset is already traded at time $t_s^{j_k}$, and 0 otherwise. The $\mathbf{Z}_k^s$'s are independent of each other for $k = 1, \ldots, p-b_s-1$, each with independent standard normal entries. Define

$$g_{ij}^s = ((\mathbf{X}_{v_s} - \mathbf{X}(s))^{\mathrm{T}}\mathbf{x}_{ij})^2 - \sum_{k=1}^{p-b_s-1} \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_k^s\mathbf{\Sigma}(t_s^{j_k}, t_s^{j_{k+1}})\mathbf{D}_k^s\mathbf{x}_{ij} = \sum_{k_1,k_2}^{p-b_s-1} h_{k_1k_2,ij}^s, \quad \text{where}$$

$$h_{k_1k_2,ij}^s = \begin{cases} \mathbf{Z}_{k_1}^{s\mathrm{T}}\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})^{1/2}\mathbf{D}_{k_1}^s\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_2}^s\mathbf{\Sigma}(t_s^{j_{k_2}}, t_s^{j_{k_2+1}})^{1/2}\mathbf{Z}_{k_2}^s, & k_1 \neq k_2; \\ (\mathbf{Z}_{k_1}^{s\mathrm{T}}\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})^{1/2}\mathbf{D}_{k_1}^s\mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_1}^s\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})\mathbf{D}_{k_1}^s\mathbf{x}_{ij}, & k_1 = k_2. \end{cases}$$

which have $E(g_{ij}^s|\mathbf{x}) = E(h_{k_1k_2,ij}^s|\mathbf{x}) = 0$, and each g_{ij}^s is uncorrelated of each other for $s \in S^j(K)$, while $h_{k_1k_2,ij}^s$ is uncorrelated of each other as long as either the index k_1 or k_2 are different, for fixed i, j and s. Then for $q \geq 2$ and $k_1 \neq k_2$,

$$\begin{aligned} E(|h_{k_1k_1,ij}^s|^q|\mathbf{x}) &= O((\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_1}^s\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})\mathbf{D}_{k_1}^s\mathbf{x}_{ij})^q) = O(n^{-q}L^{-q}(p-b_s-1)^{-q}), \\ E(|h_{k_1k_2,ij}^s|^q|\mathbf{x}) &\leq \prod_{r=1}^{2} E^{1/2}((h_{k_rk_r,ij}^s + (\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_r}^s\mathbf{\Sigma}(t_s^{j_{k_r}}, t_s^{j_{k_r+1}})\mathbf{D}_{k_r}^s\mathbf{x}_{ij}))^q|\mathbf{x}) = O(n^{-q}L^{-q}(p-b_s-1)^{-q}), \end{aligned}$$

where the last line used the result in the first line. Then we have for $q \geq 2$,

$$E(|g_{ij}^s|^q|\mathbf{x}) \leq E^{1/2}(|g_{ij}^s|^{2q}|\mathbf{x}) = O(((p-b_s-1)^{2q} \cdot n^{-2q}L^{-2q}(p-b_s-1)^{-2q})^{1/2}) = O(n^{-q}L^{-q}). \quad (7.13)$$

With this result, we can now decompose

$$\begin{aligned} R_{x,1} &= \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \sum_{s\in S^j(K)} g_{ij}^s \right| + R, \quad \text{where} \\ R &= \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \sum_{s\in S^j(K)} \sum_{k=1}^{p-b_s-1} \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_k^s\mathbf{\Sigma}(t_s^{j_k}, t_s^{j_{k+1}})\mathbf{D}_k^s\mathbf{x}_{ij} \right| = O(1/K) = o(1). \end{aligned}$$

The first term has

$$E((R_{x,1} - R)^{2q}|\mathbf{x}) = O(pL \cdot L^{2q}K^{-2q}|S^j(K)|^q \cdot n^{-2q}L^{-2q}) = O(n^{-q+1}LK^{-2q}) = O(n^{-7q/3+14/13}),$$

so that we can set $q = 1$ and through the Borel-Cantelli lemma, $R_{x,1} - R \overset{a.s.}{\to} 0$, and so $R_{x,1} \overset{a.s.}{\to} 0$ since $R = o(1)$.

For the term $R_{x,2}$, it can be bounded as

$$\begin{aligned}
R_{x,2} &= \frac{2L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K}(\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell))\right|\\
&\le \frac{2L}{C_1C_3}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\max_{0\le\ell\le K-1}\left|\sum_{r=1}^{|S^j(K)|_K}g^j_{r,\ell}\right|,\quad\text{where}\\
g^j_{r,\ell} &= (\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell)),
\end{aligned}$$

with $\mathbf{X}^j(s)$ representing the process $\mathbf{X}$ in the jth partition, and the index s goes from 0 to $|S^j(K)|$. Note that we have $E(g^j_{r,\ell}|\mathbf{x})=0$, and each $g^j_{r,\ell}$ is independent of each other for fixed j,ℓ and $r=1,\dots,|S^j(K)|_K$. For $q\ge 2$, we then have

$$\begin{aligned}
&E(|g^j_{r,\ell}|^q|\mathbf{x})\le E^{1/2}(((\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij})^{2q}|\mathbf{x})E^{1/2}((\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell)))^{2q}|\mathbf{x})\\
&\le 2^{(q-1)/2}\Big(\mathbf{E}^{1/2}\big(((\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{rK+\ell}})^{\mathrm{T}}\mathbf{x}_{ij})^{2q}|\mathbf{x}\big)+E^{1/2}\big(((\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij})^{2q}|\mathbf{x}\big)\Big)\\
&\quad\cdot E^{1/2}((\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell)))^{2q}|\mathbf{x})\\
&=O((n^{-q}L^{-q})^{1/2}+(K^qn^{-q}L^{-q})^{1/2})\cdot O((n^{-q}L^{-q})^{1/2})=O(K^{q/2}n^{-q}L^{-q}),
\end{aligned}$$

where we used the results from (7.11) and (7.13) in the last line. Then for $q\ge 1$,

$$E(R^{2q}_{x,2}|\mathbf{x})=O(pLK\cdot L^{2q}|S^j(K)|^q_KK^qn^{-2q}L^{-2q})=O(LKn^{-q+1})=O(n^{-q+68/39}),$$

so that we can set $q=3$, and through the Borel-Cantelli lemma to show that $R_{x,2}\xrightarrow{a.s.}0$. Hence we finally have $R_x\le R_{x,1}+R_{x,2}\xrightarrow{a.s.}0$. Similarly, we can show using the same set of arguments that $R^{(K)}_x\xrightarrow{a.s.}0$, so that $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}^j_3\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0$. Note that the same arguments used in the proof of this lemma can be applied to show that $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}^j_4\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0$.

For the part where $\mathbf{X}_{v_s}$ is replaced by $\mathbf{X}_{v_s}+\mathbf{J}_{v_s}-\widehat{\mathbf{J}}_{v_s}$, observe that the summations involved in the definitions of $R_\epsilon, R^{(K)}_\epsilon, R_x$ and $R^{(K)}_x$ have only constant number of terms multiplied by K when they involve cross terms $\Delta_K\mathbf{J}_{v_s}-\Delta_K\widehat{\mathbf{J}}_{v_s}$, $\mathbf{J}_{v_s}-\mathbf{J}(s)+\widehat{\mathbf{J}}_{v_s}-\widehat{\mathbf{J}}(s)$ or $\mathbf{J}_{v_{s-K}}-\mathbf{J}(s-K)+\widehat{\mathbf{J}}_{v_{s-K}}-\widehat{\mathbf{J}}(s-K)$, where $\mathbf{J}(s)$ has similar definition as $\mathbf{X}(s)$. We can use similar arguments as in the proof of Lemma 1 to show that then $R_\epsilon, R^{(K)}_\epsilon, R_x$ and $R^{(K)}_x$ are all going to 0 almost surely.

This completes the proof of the lemma. □

Lemma 3. *Let Assumptions (A1) to (A6) hold. Then for unit vectors $\mathbf{x}_{ij}$ being independent of $\{\boldsymbol{\epsilon}_s\}_{s\in S^j(1)}$*

for $i = 1, \ldots, p$ and $j = 1, \ldots, L$, we have for $\mathbf{B}_5^j, \mathbf{B}_6^j$ defined in (7.5),

$$\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{B}_5^j + \mathbf{B}_6^j)\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}} \right| \xrightarrow{a.s.} 0.$$

Suppose in the definition of (7.4), $\mathbf{X}_{v_s}$ is replaced by

$$\mathbf{X}_{v_s} + \mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s},$$

where $\mathbf{J}_{v_s}$ represents the accumulated jumps up to time v_s (see equation (3.2) and the details therein), and $\widehat{\mathbf{J}}_{v_s}$ represents the estimator of $\mathbf{J}_{v_s}$ using the wavelet method from Fan and Wang (2007) as described in Section 3.1. Then the same results in the lemma hold true if Assumptions (W1) to (W3) from Section 3.1 hold also.

Proof of Lemma 3. Similar to the arguments in (7.6), we can show that

$$\begin{aligned}
\frac{1}{4}\mathbf{x}_{ij}^{\mathrm{T}}([\mathbf{E}^+, \mathbf{E}^+]_j^{(K)} - [\mathbf{E}^-, \mathbf{E}^-]_j^{(K)})\mathbf{x}_{ij} &= \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} ((\boldsymbol{\epsilon}_{rK+\ell}^j - \boldsymbol{\epsilon}_{(r-1)K+\ell}^j)^{\mathrm{T}}\mathbf{x}_{ij})^2 \\
= \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} ((\boldsymbol{\epsilon}^j(rK+\ell) - \boldsymbol{\epsilon}^j((r-1)K+\ell) & \\
&+ (\mathbf{X}^j(rK+\ell) - \mathbf{X}_{v_{rK+\ell}^j}) + (\mathbf{X}_{v_{(r-1)K+\ell}^j} - \mathbf{X}^j((r-1)K+\ell)))^{\mathrm{T}}\mathbf{x}_{ij}\big)^2,
\end{aligned}$$

where $\boldsymbol{\epsilon}^j(s)$ is the process $\boldsymbol{\epsilon}(\cdot)$ in the jth partition, with $s = 0, \ldots, |S^j(K)|$. This implies that we have

$$\begin{aligned}
&\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \left| \frac{\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{B}_5^j + \mathbf{B}_6^j)\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}} \right| \le I_1 + I_2 + I_3 + R, \quad \text{where} \\
I_1 &= \frac{L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \left| \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} (\boldsymbol{\epsilon}^j(rK+\ell)^{\mathrm{T}}\mathbf{x}_{ij})^2 - \frac{|S^j(K)|_K}{|S^j(1)|}\sum_{r=1}^{|S^j(1)|}(\boldsymbol{\epsilon}^j(r)^{\mathrm{T}}\mathbf{x}_{ij})^2 \right|, \\
I_2 &= \frac{L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \left| \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} (\boldsymbol{\epsilon}^j((r-1)K+\ell)^{\mathrm{T}}\mathbf{x}_{ij})^2 - \frac{|S^j(K)|_K}{|S^j(1)|}\sum_{r=1}^{|S^j(1)|}(\boldsymbol{\epsilon}^j(r-1)^{\mathrm{T}}\mathbf{x}_{ij})^2 \right|, \\
I_3 &= \frac{2L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \left| \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} \boldsymbol{\epsilon}^j(rK+\ell)^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}^j((r-1)K+\ell) \right. \\
&\qquad \left. - \frac{|S^j(K)|_K}{|S^j(1)|}\sum_{r=1}^{|S^j(1)|}\boldsymbol{\epsilon}^j(r)^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}^j(r-1) \right|,
\end{aligned}$$

and R is the absolute sum of terms involving all other products, including those between $\boldsymbol{\epsilon}^j(rK+\ell)^{\mathrm{T}}\mathbf{x}_{ij}$ or $\boldsymbol{\epsilon}^j((r-1)K+\ell)^{\mathrm{T}}\mathbf{x}_{ij}$ and $(\mathbf{X}^j(rK+\ell) - \mathbf{X}_{v_{rK+\ell}^j})^{\mathrm{T}}\mathbf{x}_{ij}$ or $(\mathbf{X}^j((r-1)K+\ell) - \mathbf{X}_{v_{(r-1)K+\ell}^j})^{\mathrm{T}}\mathbf{x}_{ij}$ (terms contributed from $\mathbf{B}_6^j$ has $K=1$ and $\ell=0$). These terms can be proved to be asymptotically 0 almost surely with techniques similar to those employed in proving R_ϵ, $R_\epsilon^{(K)}$ and $R_x \xrightarrow{a.s.} 0$ in the proof of Lemma 2. For the products between $(\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{rK+\ell}})^{\mathrm{T}}\mathbf{x}_{ij}$ and $(\mathbf{X}^j((r-1)K+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij}$, they can also be proved to be asymptotically 0 almost surely using techniques similar to proving R_x and $R_x^{(K)} \xrightarrow{a.s.} 0$ in the proof of Lemma 2. Hence we have $R \xrightarrow{a.s.} 0$ and we omit the details.

We can write $\boldsymbol{\epsilon}^j(rK+\ell) = \boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{Z}^j_{r,\ell}$, where $\mathbf{Z}^j_{r,\ell}$ is independent of each other for $r=1,\ldots,|S^j(K)|_K$, $\ell=0,\ldots,K-1$ and fixed K, with independent standard normal entries. Then we can decompose $I_1 \leq I_{1,1}+I_{1,2}$, where

$$I_{1,1} = \frac{L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} g^j_{r,\ell}\right|, \quad \text{with } g^j_{r,\ell} = (\mathbf{Z}^{j\mathrm{T}}_{r,\ell}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon\mathbf{x}_{ij},$$

$$I_{1,2} = \frac{L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\frac{|S^j(K)|_K}{|S^j(1)|}\left|\sum_{r=1}^{|S^j(1)|} g^j_{r,0}\right|, \quad \text{with } g^j_{r,0} = (\mathbf{Z}^{j\mathrm{T}}_{r,0}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon\mathbf{x}_{ij}.$$

Each $g^j_{r,\ell}$ is independent of each other for different r,ℓ and fixed j and K, with $E(g^j_{r,\ell}|\mathbf{x}) = 0$ and $E(|g^j_{r,\ell}|^q|\mathbf{x}) = O((\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon\mathbf{x}_{ij})^q) = O(1)$. Hence for $q \geq 1$,

$$E(I_{1,1}^{2q}|\mathbf{x}) = O(pL\cdot L^{2q}K^{-2q}\cdot(|S^j(K)|-K+1)^q) = O(n^{-7q/39+14/13}),$$

so that with $q=12$, we have $I_{1,1} \xrightarrow{a.s.} 0$ through the Borel-Cantelli lemma. Exactly the same arguments show $I_{1,2} \xrightarrow{a.s.} 0$. Hence $I_1 \xrightarrow{a.s.} 0$. We also have $I_2 \xrightarrow{a.s.} 0$ by the same decomposition and arguments for I_1.

Finally, we can decompose $I_3 \leq I_{3,1}+I_{3,2}$, where

$$I_{3,1} = \frac{2L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K}\mathbf{Z}^{j\mathrm{T}}_{r,\ell}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{Z}^j_{r-1,\ell}\right|,$$

$$I_{3,2} = \frac{2L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{|S^j(K)|_K}{|S^j(1)|}\sum_{r=1}^{|S^j(1)|}\mathbf{Z}^{j\mathrm{T}}_{r,0}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{Z}^j_{r-1,\ell}\right|.$$

To find the order of $I_{3,1}$, we can decompose $I_{3,1} \leq I_{3,1,\text{even}} + I_{3,1,\text{odd}}$, so for instance,

$$I_{3,1,\text{even}} = \frac{2L}{C_1C_3}\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{\substack{r=1\\r\text{ even}}}^{|S^j(K)|_K} g^j_{r,\ell}\right|, \quad \text{where } g^j_{r,\ell} = \mathbf{Z}^{j\mathrm{T}}_{r,\ell}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{Z}^j_{r-1,\ell}.$$

Then each $g^j_{r,\ell}$ is independent of each other for r even and for each ℓ and fixed j, with $E(g^j_{r,\ell}|\mathbf{x}) = 0$ and for $q \geq 2$, $E(|g^j_{r,\ell}|^q|\mathbf{x}) = O((\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon \mathbf{x}_{ij})^q) = O(1)$. Hence

$$E(|I_{3,1,\text{even}}|^{2q}|\mathbf{x}) = O(pL \cdot L^{2q}K^{-2q}[|S^j(K)|_K/2 \cdot K]^q) = O(n^{-7q/39+14/13}),$$

so that at $q = 12$ we have $I_{3,1,\text{even}} \overset{a.s.}{\to} 0$. Same arguments show that $I_{3,1,\text{odd}} \overset{a.s.}{\to} 0$, and hence $I_{3,1} \overset{a.s.}{\to} 0$. We can decompose $I_{3,2}$ in the same way to show that $I_{3,2} \overset{a.s.}{\to} 0$. Therefore, we have $I_3 \overset{a.s.}{\to} 0$.

For the part where $\mathbf{X}_{v_s}$ is replaced by $\mathbf{X}_{v_s} + \mathbf{J}_{v_s} - \widehat{\mathbf{J}}_{v_s}$, arguments similar to the proof of Lemma 1 can be used to show that $R \overset{a.s.}{\to} 0$, where any summations on the dummy variable r involving the cross terms $\mathbf{J}(rK+\ell) - \widehat{\mathbf{J}}(rK+\ell) - \mathbf{J}_{v_{rK+\ell}} + \widehat{\mathbf{J}}_{v_{rK+\ell}}$ or $\mathbf{J}((r-1)K+\ell) - \widehat{\mathbf{J}}((r-1)K+\ell) - \mathbf{J}_{v_{(r-1)K+\ell}} + \widehat{\mathbf{J}}_{v_{(r-1)K+\ell}}$ has only constant number of terms.

This completes the proof of the lemma. □

Proof of Theorem 1. By the independent increment property of the diffusion process in (2.1) for $\{\mathbf{X}_t\}$, and the serial independence of $\{\boldsymbol{\epsilon}(s)\}$ by Assumption (A3), we have $\widetilde{\boldsymbol{\Sigma}}_{-j}$, and hence $\mathbf{P}_{-j}$, are independent of $\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ for each $j = 1, \ldots, L$. Writing $\mathbf{p}_{ji}$ as the ith eigenvector of $\mathbf{P}_{-j}$, we can then apply Lemma 1, 2 and 3 with $\mathbf{x}_i = \mathbf{p}_{ji}$ to get

$$\begin{aligned}
\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}} \left| \frac{\mathbf{p}_{ji}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1},\tau_j)\mathbf{p}_{ji}}{\mathbf{p}_{ji}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{p}_{ji}} - 1 \right| &\leq \max_{\substack{i=1,\ldots,p\\1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}}\mathbf{B}_1\mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{\ell-1},\tau_\ell)\mathbf{x}_i} - 1 \right| + \max_{\substack{i=1,\ldots,p\\1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}}\mathbf{B}_2\mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{\ell-1},\tau_\ell)\mathbf{x}_i} \right| \\
&\quad + \max_{\substack{i=1,\ldots,p\\1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}}(\mathbf{B}_3+\mathbf{B}_4)\mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{\ell-1},\tau_\ell)\mathbf{x}_i} \right| + \max_{\substack{i=1,\ldots,p\\1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}}(\mathbf{B}_5+\mathbf{B}_6)\mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{\ell-1},\tau_\ell)\mathbf{x}_i} \right| \\
&\overset{a.s.}{\to} 0.
\end{aligned}$$

Finally, note that

$$\begin{aligned}
&\max_{1\leq j\leq L} \left\|\widehat{\boldsymbol{\Sigma}}(\tau_{j-1},\tau_j)\boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1},\tau_j)^{-1} - \mathbf{I}_p\right\| \\
&= \max_{1\leq j\leq L} \left\|\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1},\tau_j)\mathbf{P}_{-j})\text{diag}^{-1}(\mathbf{P}_{-j}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{P}_{-j}) - \mathbf{I}_p\right\| \\
&= \max_{\substack{i=1,\ldots,p\\1\leq j\leq L}} \left| \frac{\mathbf{p}_{ji}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1},\tau_j)\mathbf{p}_{ji}}{\mathbf{p}_{ji}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{p}_{ji}} - 1 \right|,
\end{aligned}$$

which goes to 0 almost surely. This completes the proof of the theorem. □

Proof of Theorem 3. Note that by the last parts of Lemma 1, 2 and 3 where jumps are involved, the arguments in the proof of Theorem 1 hold exactly as they are. The arguments before Corollary 2 follows also. This completes the proof of Theorem 3. □

Proof of Theorem 4. Define $\mathbf{D}_j = \text{diag}(\mathbf{P}_{-j}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{P}_{-j})$ and $\widetilde{\mathbf{D}}_j = \text{diag}(\mathbf{P}_{-j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1},\tau_j)\mathbf{P}_{-j})$. Then with $\mathbf{R} = \sum_{j=1}^{L} \mathbf{P}_{-j}(\widetilde{\mathbf{D}}_j \mathbf{D}_j^{-1} - \mathbf{I}_p)\mathbf{D}_j \mathbf{P}_{-j}^{\mathrm{T}}$,

$$\begin{aligned}
\widehat{\mathbf{\Sigma}}(0,1)^{-1} &= \Big(\sum_{j=1}^{L} \mathbf{P}_{-j}\widetilde{\mathbf{D}}_j \mathbf{P}_{-j}^{\mathrm{T}}\Big)^{-1} = \Big(\sum_{j=1}^{L} \mathbf{P}_{-j}(\widetilde{\mathbf{D}}_j \mathbf{D}_j^{-1} - \mathbf{I}_p)\mathbf{D}_j \mathbf{P}_{-j}^{\mathrm{T}} + \sum_{j=1}^{L} \mathbf{P}_{-j}\mathbf{D}_j \mathbf{P}_{-j}^{\mathrm{T}}\Big)^{-1} \\
&= (\mathbf{I}_p + \mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{R})^{-1}\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1} \\
&= \mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1} + \sum_{k\geq 1}\big(-\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{R}\big)^k \mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1},
\end{aligned}$$

where the Neumann's series expansion in the last line is valid since

$$\begin{aligned}
\sum_{k\geq 0}\big\|\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|^k \big\|\mathbf{R}\big\|^k &\leq 1 + \sum_{k\geq 1}\frac{\big\|\mathbf{R}\big\|^k}{\lambda_{\min}^k(\mathbf{\Sigma}_{\text{Ideal}}(0,1))} \\
&\leq 1 + \sum_{k\geq 1}\frac{L^k \max_{1\leq j\leq L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p\big\|^k \max_{1\leq j\leq L}\big\|\mathbf{\Sigma}(\tau_{j-1},\tau_j)\big\|^k}{L^k \min_{1\leq j\leq L}\lambda_{\min}^k(\mathbf{\Sigma}(\tau_{j-1},\tau_j))} \\
&\leq 1 + \sum_{k\geq 1}\Big(\frac{C_2C_4}{C_1C_3}\max_{1\leq j\leq L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p\big\|\Big)^k \\
&\overset{a.s.}{\to} 1 < \infty,
\end{aligned}$$

where C_1, C_2, C_3, C_4 are constants in Assumptions (A2) and (A4), and the last line follows from the asymptotic almost sure convergence of $\max_{1\leq j\leq L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p\big\|$ to 0 in Theorem 1. This implies that, asymptotically almost surely,

$$\begin{aligned}
\big\|\widehat{\mathbf{\Sigma}}(0,1)^{-1} - \mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\| &\leq \lambda_{\max}(\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1})\sum_{k\geq 1}\frac{\big\|\mathbf{R}\big\|^k}{\lambda_{\min}^k(\mathbf{\Sigma}_{\text{Ideal}}(0,1))} \\
&\leq \frac{1}{L\min_{1\leq j\leq L}\lambda_{\min}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}\cdot\sum_{k\geq 1}\Big(\frac{C_2C_4}{C_1C_3}\max_{1\leq j\leq L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p\big\|\Big)^k \\
&\leq \frac{1}{C_1C_3}\sum_{k\geq 1}\Big(\frac{C_2C_4}{C_1C_3}\max_{1\leq j\leq L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p\big\|\Big)^k \overset{a.s.}{\to} 0. \qquad (7.14)
\end{aligned}$$

From the formulae of $\widehat{\mathbf{w}}_{\text{opt}}$ and $\mathbf{w}_{\text{Ideal}}$ given at the beginning of Section 3.2,

$$\text{Eff}(\mathbf{w}_{\text{Ideal}}, \widehat{\mathbf{w}}_{\text{opt}}) = \frac{\mathbf{1}_p^{\mathrm{T}}\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{\Sigma}(0,1)\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\mathbf{\Sigma}}(0,1)^{-1}\mathbf{\Sigma}(0,1)\widehat{\mathbf{\Sigma}}(0,1)^{-1}\mathbf{1}_p}\cdot\Big(\frac{\mathbf{1}_p^{\mathrm{T}}\widehat{\mathbf{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}\Big)^2.$$

With (7.14), the first term of the product on the right hand side above can be decomposed into $1 + R_1 + R_2$, where

$$R_1 = \frac{\mathbf{1}_p^{\mathrm{T}}(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1} - \widehat{\boldsymbol{\Sigma}}(0,1)^{-1})\boldsymbol{\Sigma}(0,1)\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p},$$
$$R_2 = \frac{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1} - \widehat{\boldsymbol{\Sigma}}(0,1)^{-1})\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p},$$

with, asymptotically,

$$\|R_1\| \leq \frac{\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\| \cdot \frac{C_2C_4}{C_1C_3}\lambda_{\max}^2(\widehat{\boldsymbol{\Sigma}}(0,1))}{C_1C_3} \xrightarrow{a.s.} 0,$$

since we have (7.14) and that by Theorem 1 and 3, asymptotically almost surely, $\lambda_{\max}^2(\widehat{\boldsymbol{\Sigma}}(0,1)) \leq C_2C_4$. Similarly, we can prove that asymptotically, $R_2 \xrightarrow{a.s.} 0$, as well as

$$\frac{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p} \xrightarrow{a.s.} 1.$$

This completes the proof of the theorem. □

Proof of Theorem 5. Define $\mathbf{e}_i$ to be the unit vector with 1 on the ith position and 0 elsewhere, and $\|\mathbf{A}\|_1 = \max_j \sum_i |a_{ij}|$ the L_1 norm of a matrix $\mathbf{A}$. For some $i = 1, \ldots, p$, using the same notations as in the proof of Theorem 4,

$$\begin{aligned}
p^{1/2}\|\widehat{\mathbf{w}}_{\text{opt}}\|_{\max} &= \frac{p^{1/2}|\mathbf{e}_i^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p|}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p} \leq \frac{p^{1/2}\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\|_1}{p\lambda_{\min}(\widehat{\boldsymbol{\Sigma}}(0,1)^{-1})} \leq \frac{p^{1/2}\cdot p^{1/2}/\lambda_{\min}(\widehat{\boldsymbol{\Sigma}}(0,1))}{p/\lambda_{\max}(\widehat{\boldsymbol{\Sigma}}(0,1))} \\
&\leq \frac{\sum_{j=1}^{L}\lambda_{\max}(\widetilde{\mathbf{D}}_j)}{\sum_{j=1}^{L}\lambda_{\min}(\widetilde{\mathbf{D}}_j)} \\
&\leq \frac{L\max_{1\leq j\leq L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p)\lambda_{\max}(\mathbf{D}_j) + \sum_{j=1}^{L}\lambda_{\max}(\mathbf{D}_j)}{L\min_{1\leq j\leq L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p)\lambda_{\min}(\mathbf{D}_j) + \sum_{j=1}^{L}\lambda_{\min}(\mathbf{D}_j)} \\
&\leq \frac{C_2C_4\max_{1\leq j\leq L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p) + \sum_{j=1}^{L}\lambda_{\max}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}{C_1C_3\min_{1\leq j\leq L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p) + \sum_{j=1}^{L}\lambda_{\min}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))} \\
&= \frac{C_2C_4\max_{1\leq j\leq L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p) + \lambda_{\max}\Big(\sum_{j=1}^{L}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\Big)}{C_1C_3\min_{1\leq j\leq L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p) + \lambda_{\min}\Big(\sum_{j=1}^{L}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\Big)} \xrightarrow{a.s.} \mathrm{Cond}(\boldsymbol{\Sigma}(0,1)),
\end{aligned}$$

where the last line follows from Assumption (A5) that each $\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)$ has the same set of eigenvectors and the same order of eigenvalues, and that by Theorem 1 and Theorem 3 both

$$\frac{\max_{1\leq j\leq L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p)}{\lambda_{\min}(\boldsymbol{\Sigma}(0,1))}, \quad \frac{\min_{1\leq j\leq L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1} - \mathbf{I}_p)}{\lambda_{\min}(\boldsymbol{\Sigma}(0,1))}$$

go to 0 asymptotically almost surely.

For the actual risk upper bound, consider the decomposition $pR(\widetilde{\mathbf{w}}_{\text{opt}}) = I_1 + I_2 + I_3$, where

$$\begin{aligned}
I_1 &= \frac{p\mathbf{1}_p^{\mathrm{T}}(\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{(\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p)^2}, \\
I_2 &= \frac{p\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)(\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})\mathbf{1}_p}{(\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p)^2}, \\
I_3 &= \frac{p\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{(\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p)^2}.
\end{aligned}$$

By (7.14),

$$|I_1| \le \frac{p^2\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\| \cdot C_2C_4 \cdot \big(\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\| + \frac{1}{C_1C_3}\big)}{p^2\big(\lambda_{\min}(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}) - \big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|\big)^2} \overset{a.s.}{\to} 0.$$

Similarly, $|I_2| \overset{a.s.}{\to} 0$. For I_3, by (7.14),

$$\begin{aligned}
|I_3| &\le \frac{p^2\lambda_{\max}^2(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})\lambda_{\max}(\boldsymbol{\Sigma}(0,1))}{p^2\big(\lambda_{\min}(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}) - \big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1} - \boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|\big)^2} \\
&\overset{a.s.}{\to} \frac{\lambda_{\max}^2(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1))}{\lambda_{\min}^2(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1))}\lambda_{\max}(\boldsymbol{\Sigma}(0,1)) \\
&\le \left(\frac{\sum_{j=1}^L \lambda_{\max}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}{\sum_{j=1}^L \lambda_{\min}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}\right)^2 \lambda_{\max}(\boldsymbol{\Sigma}(0,1)) \\
&= \left(\frac{\lambda_{\max}(\sum_{j=1}^L \boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}{\lambda_{\min}(\sum_{j=1}^L \boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}\right)^2 \lambda_{\max}(\boldsymbol{\Sigma}(0,1)) = \mathrm{Cond}^2(\boldsymbol{\Sigma}(0,1))\lambda_{\max}(\boldsymbol{\Sigma}(0,1)),
\end{aligned}$$

where the last line follows from Assumption (A5). This completes the proof of the theorem. □

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